
From Static Constraints to Dynamic Adaptation: Sample-Level Constraint Relaxation for Offline-to-Online Reinforcement Learning

Lipeng Zu¹ Yu Qian¹ Shayok Chakraborty¹ Xiaonan Zhang¹

Abstract

Offline-to-online reinforcement learning (O2O RL) faces a central challenge between retaining offline conservatism and adapting to online feedback under distribution shift. This challenge arises because data behavior evolves during fine-tuning, rendering data origin a misleading basis for constraint handling and thereby leading to objective–data mismatch. We therefore propose Dynamic Alignment for RELaxation (DARE), a distribution-aware framework for sample-level constraint relaxation based on the behavioral consistency with a behavior model. To our knowledge, DARE is the first to condition constraint relaxation on behavioral consistency via a posterior-induced exchange mechanism, moving beyond a binary offline/online data distinction. Importantly, DARE requires only per-sample behavioral alignment, enabling instantiation on top of many offline algorithms with flexible choices of behavior models and fine-tuning objectives. We provide a theoretical analysis showing that behavior-based sample exchange consistently improves the distinction between offline-like and online-like subsets. Experiments on D4RL demonstrate that DARE consistently improves fine-tuning stability and achieves superior final performance over strong offline-to-online baselines. (The code is publicly available at <https://github.com/lpzu/DARE>.)

1. Introduction

Offline-to-online reinforcement learning (O2O RL) aims to combine the efficiency of offline pretraining with the adaptability of online fine-tuning (Xie et al., 2021; Zhang

& Zanette, 2023). By initializing policies from offline dataset (Fujimoto et al., 2019; Kumar et al., 2020; Uehara et al., 2024) and refining them through limited online interactions (Mnih et al., 2015; Song et al., 2023; Alonso et al., 2024), O2O RL provides a practical pathway toward scalable and safe RL in complex tasks (Lee et al., 2022; He et al., 2025). However, offline pretraining typically relies on conservative constraints to ensure stability in the absence of online feedback (Ball et al., 2023). Once the agent enters online fine-tuning, a distribution shift naturally arises because the evolving policy induces behaviors that differ from those represented in the offline dataset (An et al., 2021; Yang et al., 2022). As a result, conservative constraints can conflict with the need for exploration and adaptation (Yu & Zhang, 2023). This inherent discrepancy lies at the central challenge of the O2O setting.

To cope with this fundamental challenge, most existing O2O approaches regulate policy updates or value estimation to preserve stability (Wu et al., 2022; Zhang et al., 2023). In practice, the regulation is achieved via explicitly removing conservative constraints inherited from offline training or incorporating them through soft transition mechanisms during online fine-tuning (Wang et al., 2023; Xu et al., 2025a). Despite these design choices, these methods typically apply uniform training objectives across samples from different data origins (Luo et al., 2024; Mao et al., 2024). This uniformity thereby obfuscates the distinction between samples that are meant to anchor conservative calibration and those that are meant to support online adaptation.

A seemingly natural response is to treat data differently based on its origin, followed by applying different training objectives accordingly (Zheng et al., 2023; Shin et al., 2025). This strategy uses data origin as the basis for assigning samples to different constraint rules. However, the distribution shift that matters in O2O RL is characterized by behavior relative to the state–action support of the offline dataset, which cannot be inferred from collection origin alone. As fine-tuning proceeds, both the offline dataset and newly collected online data can contain a mixture of offline-like and online-like behaviors. Consequently, separating data by origin still leads to the same objective–data misalignment as uniform training. Such observation motivates a sample-level

¹Department of Computer Science, Florida State University, Tallahassee, FL, USA. Correspondence to: Xiaonan Zhang <xzhang14@fsu.edu>.

approach to constraint relaxation that relies on behavioral consistency rather than data origin.

Building on the above analysis, we highlight three design requirements for constraint relaxation decision: (i) the decision criterion is behavior-driven rather than origin-driven; (ii) the decision granularity operates at the sample level to reflect heterogeneous roles within mixed data; and (iii) the decision rule is explicit and consistent within a well-defined formulation. To satisfy these requirements, we propose Dynamic Alignment for RELaxation (DARE), a distribution-aware framework for sample-level constraint handling. DARE evaluates sample-level behavioral alignment through a Bayes-optimal posterior. The posterior induced by the quadratic structure determines which samples can be exchanged between conservative and adaptive regularization, yielding a unified exchange rule for constraint-relaxation decisions.

We theoretically derive the posterior-induced exchange mechanism and demonstrate that it enhances the separability between offline-like and online-like behaviors. Empirically, we instantiate DARE within Cal-QL (Nakamoto et al., 2023) and IQL (Kostrikov et al., 2022), achieving consistent performance gains over the corresponding baselines on standard benchmarks. Additional experiments and ablations on different instantiations further validate the effectiveness of the DARE framework.

2. Related Work

Constraint Handling in O2O RL. Offline-to-online RL aims to combine reliable value estimation from offline data with adaptive policy learning during online fine-tuning under distributional shift (He et al., 2025; Li et al., 2025). Some approaches, such as FamO2O (Wang et al., 2023) and Uni-RL (Xu et al., 2025a), adopt a uniform objective formulation throughout training to avoid abrupt changes in learning signals. Other methods focus on stabilizing training dynamics: Cal-QL (Nakamoto et al., 2023) calibrates Q-values during offline training, and several online approaches (Zhang et al., 2024a; Feng et al., 2024; Ball et al., 2023) further stabilize value learning. Some recent work addresses the instability of ensemble Q-functions after offline-to-online transfer (Guo et al., 2026; Hu et al., 2024), where critic disagreement or biased value aggregation can be amplified by newly collected online data. However, these methods primarily keep stabilization via uniform training objective, without explicitly distinguishing offline and online data. In contrast, other approaches explicitly separate optimization objectives by data origin (Zheng et al., 2023). For example, OPT (Shin et al., 2025) retrain the critic with online-style objectives applied based on data origin and accelerates adaptation by increasing the update-to-data (UTD) ratio. Nevertheless, such separations are usually defined at

the level of data origin, yielding coarse-grained assignments that overlook sample-level behavioral heterogeneity.

Behavior Modeling and Distribution Alignment. Behavior modeling has been widely explored as an approach to providing a reference for aligning learning dynamics with offline data. To model behavioral imitation in a generative manner, diffusion-based behavior modeling has been explored as a generative prior over actions or trajectories (Janer et al., 2022; Chen et al., 2023; Lu et al., 2023a). Within this line of work, CEP (Lu et al., 2023b) approximates energy-based guidance through contrastive learning, primarily focusing on offline behavior generation, and EDIS (Liu et al., 2024) extends energy-guided diffusion to the O2O setting by steering action sampling during fine-tuning. In contrast to approaches that explicitly model behavior (Rigter et al., 2024), imitation-regularized methods such as AWAC (Nair et al., 2020) apply advantage-weighted imitation to keep policy updates aligned with offline behaviors, and PEX (Zhang et al., 2023) preserves offline-consistent behaviors during online adaptation through an offline-trained policy. Separately, Off2On (Lee et al., 2022) interpolates between conservative offline pretraining and exploratory online fine-tuning via replay reweighting. Across these methods, behavioral components are predominantly deployed for regularization or generation, without being explicitly designed to guide sample-level constraint handling.

Unlike prior work that assigns objectives by training phase or data origin, we leverage behavioral consistency relative to a learned behavior model to make sample-level constraint-handling decisions.

3. Preliminaries

3.1. Reinforcement Learning

RL is a framework in which an agent learns to maximize cumulative rewards by interacting with an environment (Mnih et al., 2013; Van Hasselt et al., 2016). The problem is modeled as a Markov Decision Process (MDP), defined by a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P is the transition probability, R is the reward function, and $\gamma \in [0, 1)$ is the discount factor.

3.2. Offline-to-online RL

Offline-to-online (O2O) RL commonly follows a two-phase training paradigm (Hu et al., 2024; Zheng et al., 2025): (i) *offline pretraining*, where a policy is learned from a fixed dataset $\mathcal{D}_{\text{off}}$ without interacting with the environment; and (ii) *online fine-tuning*, where the pre-trained policy interacts with the environment, and the newly collected transitions are stored in an online replay buffer $\mathcal{D}_{\text{on}}$. During online fine-tuning, each update uses a mini-batch b mixing samples from both buffers, which is $b = b_{\text{off}} \cup b_{\text{on}}$.

3.3. Constraints in O2O RL

Conservative constraints are widely used in offline RL to stabilize value estimation under distribution shift. During the offline-to-online transition, constraint handling often involves a trade-off between enabling online adaptation and maintaining stable policy improvement. In many algorithms, the constraint is encoded through the learning objective. We outline two representative classes below

- **Explicit conservatism.** Conservative Q-Learning (CQL) explicitly regularizes the Q-function to discourage overestimation on actions that are weakly supported by the data distribution (Kumar et al., 2020). In general, the explicit conservatism can be implemented by augmenting the Bellman-error objective with a regularizer $\mathcal{R}$

$$\mathcal{L}_{\text{CQL}}(Q) = \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \left[\left(Q(s,a) - \hat{B}^\pi \hat{Q}_{\text{target}}(s,a) \right)^2 \right] + \alpha \mathcal{R},$$

where α controls the strength of conservatism.

- **Implicit conservatism.** Implicit Q-Learning (IQL) induces conservatism implicitly through expectile-based value learning, discouraging the exploitation of unseen actions (Kostrikov et al., 2022). Briefly, IQL first learns an implicit value function through expectile regression

$$\mathcal{L}_{\text{IQL}}(V) = \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[L_\tau^2 \left(\hat{Q}_{\text{target}}(s,a) - V(s) \right) \right],$$

where $L_\tau^2(u) = |\tau - \mathbb{1}(u < 0)|u^2$ is the expectile regression loss. The learned value function is then used to update the Q-function and corresponding actor function.

CQL and IQL can both be viewed as mechanisms for enforcing conservatism within value-based policy optimization. In this paper, we use them to demonstrate and validate our approach. More broadly, the proposed approach is generic and can be integrated into a wide range of O2O RL.

4. Motivation

Most existing O2O RL methods fine-tune on mixed offline and online data using a single optimization objective. As the behavior distribution changes during fine-tuning, a unified objective struggles to balance training stability with online adaptation. A straightforward approach is to employ different optimization objectives for offline and online data respectively, thereby decoupling conservative calibration from online improvement. However, this approach still relies on the assumption that offline and online data are behaviorally separable. To examine the validity of this assumption under realistic fine-tuning, we conduct a diagnostic experiment.

We study the HalfCheetah environment (Towers et al., 2025) under the O2O setting by collecting a 100k-size online replay buffer. We then compare this buffer to the offline

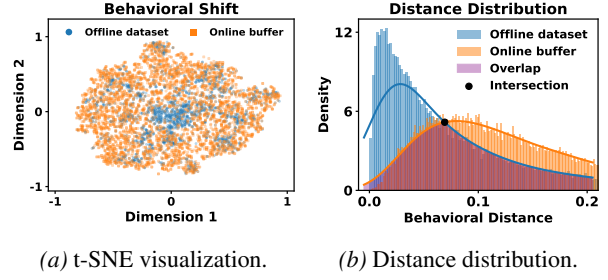


Figure 1. Characteristics of behavior shift in fine-tuning.

dataset using an offline behavior imitation model, which serves as a behavioral reference for cross-dataset comparison. As shown in Fig. 1a, offline samples concentrate in a central region of the learned representation space, whereas online samples are more dispersed. Notably, there is a non-trivial overlap between offline and online samples, demonstrating that offline and online data are not strictly separable in the behavior space.

To quantify the overlap between offline and online data, for each sample, we compute its behavioral distance to the offline imitation model. The resulting distributions in Fig. 1b exhibit a clear intersection between offline and online samples. Rather than forming disjoint groups, the two datasets occupy overlapping regions along a continuous behavioral axis. As a result, assigning optimization objectives solely based on data origin inevitably misaligns with the behavioral roles of individual samples. In particular, samples near the overlap region cannot be reliably assigned a fixed constraint role, since they may support either conservative calibration or online adaptation depending on their behavioral alignment. This observation motivates a sample-level mechanism that dynamically reassigns constraint roles, rather than enforcing static and origin-based distinctions.

5. DARE: Dynamic Alignment for RELaxation

We propose DARE, an O2O RL framework built around a posterior-induced exchange mechanism for sample-level constraint handling. The exchange mechanism leverages behavioral alignment to infer whether each sample is offline-like or online-like, and reassigns constraint roles accordingly. DARE preserves conservative constraints for offline-like samples while selectively relaxing them for samples exhibiting online-like behavior. We further show that the exchange mechanism admits a Bayes-optimal posterior interpretation and provably improves offline–online behavioral separability. All formal proofs are deferred to **Appendix A**.

5.1. Bayes–Optimal Posterior

We consider samples drawn from an offline dataset and an online replay buffer, indexed by a binary class label

$C \in \{0, 1\}$. Let $(s, a) \in \mathcal{S} \times \mathcal{A}$ denote a state–action pair, and let π_b be a reference behavior model. We define a scalar behavioral deviation statistic

$$d(s, a) := \text{dist}(a, a_b(s)), \quad (1)$$

where $\text{dist}(\cdot, \cdot)$ is a deviation measure that quantifies behavioral discrepancy and $a_b(s)$ denotes a reference action induced by π_b . We instantiate $\text{dist}(\cdot, \cdot)$ as the ℓ_2 distance.

Let $X := A \mid S$ denote the state-conditioned action variable from a behavioral perspective. We model the source discrimination being governed by behavioral deviation relative to the reference model, formalized through the following likelihood–ratio structure.

Theorem 1. *Suppose the class-conditional densities admit a likelihood–ratio representation through d ; that is, there exists a measurable function $r : \mathbb{R} \rightarrow (0, \infty)$ such that*

$$\frac{p(X = x \mid C = 0)}{p(X = x \mid C = 1)} = r(d(s, a)).$$

It follows that d is sufficient for Bayes–optimal posterior inference between the two classes, i.e. there exists a measurable function $g : \mathbb{R} \rightarrow [0, 1]$ such that

$$\mathbb{P}(C = 0 \mid X = x) = g(d(s, a)).$$

Thm. 1 shows that the source-discrimination problem can be reduced from the high-dimensional behavioral space to the scalar statistic $D = d(s, a)$. Specifically, the $r(\cdot)$ in the assumption relates the likelihood ratio to the scalar d , whereas the $g(\cdot)$ in the conclusion represents the posterior as a function of the same statistic. Since D is a measurable statistic of the variable (S, A) , it induces a sub- σ -algebra of the full state–action space

$$\sigma(D) = \sigma_b(S, A) \subseteq \sigma(S, A),$$

where $\sigma_b(S, A)$ denotes the behavior-relevant sub- σ -algebra retained by the scalar statistic D . Thus, function g in Thm. 1 is given by

$$g(d(s, a)) = \mathbb{P}(C = 0 \mid D = d). \quad (2)$$

To obtain a tractable posterior form, we further model the class–conditional distribution of D as Gaussian

$$D \mid C = c \sim \mathcal{N}(\mu_c, \sigma_c^2), \quad c \in \{0, 1\}.$$

Let $\rho_c := \mathbb{P}(C = c)$ denote the class priors, with $\rho_0, \rho_1 > 0$ and $\rho_0 + \rho_1 = 1$. By Bayes’ rule, the posterior probability in Eq. 2 admits the closed-form expression

$$\begin{aligned} \mathbb{P}(C = 0 \mid D = d) &= \frac{\rho_0 p(d \mid C = 0)}{\rho_0 p(d \mid C = 0) + \rho_1 p(d \mid C = 1)} \\ &= \frac{1}{1 + \frac{\rho_1 \sigma_0}{\rho_0 \sigma_1} \exp\left(-\frac{(d - \mu_1)^2}{2\sigma_1^2} + \frac{(d - \mu_0)^2}{2\sigma_0^2}\right)}. \end{aligned} \quad (3)$$

This posterior form serves as the basis for deriving symmetry-induced thresholds and the exchange rules designed in the subsequent sections.

Remark 1. Although the distance statistic d is nonnegative, approximating its class-conditional distribution as Gaussian is reasonable as a local approximation. For certain positive-support distributions (e.g., Gamma), the log-density admits a second-order Taylor expansion around the mode, yielding a locally quadratic form in d . Since Gaussian log-densities are quadratic, this provides intuition for using a Gaussian approximation in the neighborhood of high-probability regions. A more detailed discussion of this assumption is provided in **Appendix B**.

5.2. Symmetry-Induced Thresholds

We model the behavioral statistic D using a two-class Gaussian mixture, which allows us to compute the posterior probability via the Bayes’ rule. Although the posterior takes a logistic form, its qualitative behavior is entirely determined by the quadratic term in the likelihood ratio. Specifically, the posterior in Eq. 3 can be re-written as

$$\mathbb{P}(C = 0 \mid D = d) = \frac{1}{1 + \alpha \exp(\ell(d))}, \quad (4)$$

where $\alpha = \frac{\rho_1 \sigma_0}{\rho_0 \sigma_1} > 0$ and

$$\ell(d) = -\frac{(d - \mu_1)^2}{2\sigma_1^2} + \frac{(d - \mu_0)^2}{2\sigma_0^2} = \ell_1 d^2 + \ell_2 d + \ell_3.$$

Since the logistic function is strictly monotonic, the monotonicity and extremal behavior of the posterior in Eq. 4 are fully determined by the quadratic function $\ell(d)$. When $\sigma_0 \neq \sigma_1$, $\ell(d)$ admits a unique stationary point

$$d^* = -\frac{\ell_2}{2\ell_1} = \frac{\mu_1/\sigma_1^2 - \mu_0/\sigma_0^2}{1/\sigma_1^2 - 1/\sigma_0^2}. \quad (5)$$

Theorem 2. *The point d^* in Eq. 5 satisfies $d^* \notin (\mu_0, \mu_1)$.*

As a consequence, the posterior $\mathbb{P}(C = 0 \mid D = d)$ remains strictly monotonic throughout the interval (μ_0, μ_1) , which preserves the ordering of samples between the two class means. This property implies that any threshold chosen within this interval therefore yields a consistent separation. We therefore adopt their midpoint τ as a symmetric and parameter-free central reference threshold.

Outside the central monotonic interval, we introduce a second threshold τ_{side} , the reflection of τ about d^* , since $\ell(d)$ is a quadratic symmetric around d^* . This construction yields two thresholds that attain the same posterior value, thereby accounting for the side region associated with the stationary point. Formally, the two thresholds are defined as

$$\tau := \frac{\mu_0 + \mu_1}{2}, \quad \tau_{\text{side}} := 2d^* - \tau. \quad (6)$$

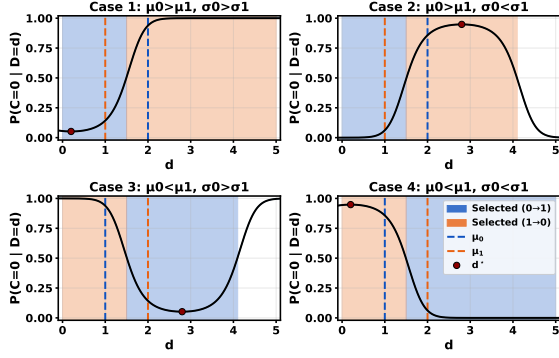


Figure 2. Posterior $\mathbb{P}(C = 0 \mid D = d)$ under four configurations of class means and standard deviations. Dashed lines denote the class means, the red dot indicates the stationary point, and shaded regions show samples selected by the proposed exchange rule.

5.3. Exchange Rule

With the thresholds τ and τ_{side} defined in Eq. 6, we now specify how posterior structure induces constraint exchange between offline and online behaviors. As illustrated in Fig. 2, the posterior in Eq. 3 exhibits four distinct qualitative patterns, depending on the relative ordering of the class means (μ_0, μ_1) and variances (σ_0, σ_1) .

To capture the orientation of the posterior in a compact form, we introduce the direction signs

$$\delta_\mu := \text{sign}(\mu_0 - \mu_1), \quad \delta_\sigma := \text{sign}(\sigma_0 - \sigma_1). \quad (7)$$

Here, δ_μ encodes the ordering of the class means, while δ_σ captures the direction of variance-induced curvature in the posterior.

Using the sign variables in Eq. 7, we express the exchange rule in a unified form as follows:

An element with projection value d from class $C = 0$ is exchanged to class $C = 1$ if and only if

$$\delta_\mu \delta_\sigma (d - \tau_{\text{side}}) > 0 \quad \text{and} \quad \delta_\mu (d - \tau) < 0.$$

An element with projection value d from class $C = 1$ is exchanged to class $C = 0$ if and only if

$$\delta_\mu \delta_\sigma (d - \tau_{\text{side}}) > 0 \quad \text{and} \quad \delta_\mu (d - \tau) > 0.$$

The two conditions in the exchange rule play complementary roles. The term $\delta_\mu (d - \tau)$ determines on which side of the mean midpoint a sample lies, indicating whether it is closer to the opposite behavioral class. The term $\delta_\mu \delta_\sigma (d - \tau_{\text{side}})$ identifies the variance-dominated side region induced by the stationary point d^* , where the posterior curve deviates from simple mean-based separation. Exchanges are triggered only when both conditions are satisfied, ensuring that constraint reassignment occurs exclusively for samples that

are both behaviorally ambiguous and located in the side region of the posterior curve. Since the rule is evaluated with support $d \in \mathbb{R}_{\geq 0}$, any induced region with $d < 0$ is inactive and cannot trigger exchanges.

Proposition 1. *The sign-based exchange rule is equivalent to the four case-by-case conditions depicted in Fig. 2.*

5.4. Exchange Operation

The exchange rule in Sec. 5.3 determines the candidate samples that should be reassigned between the two constraint roles. We then specify the corresponding exchange operation by separating the estimation of buffer-level exchange parameters from the sample-wise reassignment decision.

Buffer Level Statistics. The quantities required by the rule, including δ_μ , δ_σ , τ , and τ_{side} , are induced by the buffer-level statistics $(\mu_0, \mu_1, \sigma_0, \sigma_1)$ through Eqs. 5–7. This tuple is periodically updated from a reference set sampled from the offline and online buffers, allowing the exchange parameters to reflect the buffer-level behavior-distance distribution.

Mini-batch Level Operation. Given these buffer-level exchange parameters, the exchange is applied at the mini-batch level. Let b_{off} and b_{on} denote the offline and online mini-batches, respectively. The exchange rule in Sec. 5.3 is then used to identify two candidate pools: an offline-to-online pool $\mathcal{P}_{\text{off} \rightarrow \text{on}}$ and an online-to-offline pool $\mathcal{P}_{\text{on} \rightarrow \text{off}}$. To keep the exchange operation balanced, we swap the same number of samples in the two directions

$$K = \min(|\mathcal{P}_{\text{off} \rightarrow \text{on}}|, |\mathcal{P}_{\text{on} \rightarrow \text{off}}|). \quad (8)$$

When one candidate pool contains more than K samples, only K samples are selected from that pool, either randomly or according to the ordering of the behavioral distance d . Notably, no additional exchange criterion is introduced at this stage, because samples in the candidate pools have already been identified by the exchange rule.

5.5. Theoretical Analysis

We analyze how exchanging samples between classes affects first-order statistics and distributional separability as characterized by the exchange rule.

Let $\mathcal{B}_0 := \{d_1^{(0)}, \dots, d_{n_0}^{(0)}\}$ and $\mathcal{B}_1 := \{d_1^{(1)}, \dots, d_{n_1}^{(1)}\}$ be two finite sets of projected values with sizes n_0, n_1 . Select $x \in \mathcal{B}_0$ and $y \in \mathcal{B}_1$ such that $\delta_\mu (x - \tau) < 0$ and $\delta_\mu (y - \tau) > 0$ for exchange between the two sets.

Theorem 3. *Let μ'_0, μ'_1 denote the empirical means after the exchange. Each exchange strictly increases the absolute mean gap*

$$|\Delta'| = |\mu'_0 - \mu'_1| > |\Delta| = |\mu_0 - \mu_1|.$$

Thm. 3 establishes that the exchange procedure is mean-separation enhancing at the first-order level. To understand how the exchange reshapes the class-conditional distributions, we first characterize its effect on the empirical cumulative distribution functions (CDFs).

Lemma 1. *Consider the empirical CDF*

$$F_c(t) = \frac{1}{n_c} \sum_{i=1}^{n_c} \mathbf{1}\{d_i^{(c)} \leq t\}, \quad c \in \{0, 1\}.$$

Suppose a single element with value x in class c is swapped with an element with value y from the opposite class. Let F'_c denote the resulting empirical CDF after the swap. Then, for every $t \in \mathbb{R}$,

$$F'_c(t) = F_c(t) - (-1)^c \delta_\mu \frac{1}{n_c} \mathbf{1}\{\delta_\mu x \leq \delta_\mu t < \delta_\mu y\}.$$

We next consider the $H\Delta H$ -divergence introduced by Ben-David et al. (2010), instantiated with the one-dimensional threshold hypothesis class

$$H_{\text{thr}} = \{h_t(z) = \mathbf{1}\{z \leq t\} : t \in \mathbb{R}\}.$$

For two distributions P and Q over $Z \in \mathbb{R}$, the $H_{\text{thr}}\Delta H_{\text{thr}}$ -divergence is defined as $d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q)$.

Corollary 1. *Define $\Delta F(t) := F_0(t) - F_1(t)$, $\Delta F_{\max} := \sup_{t \in \mathbb{R}} \Delta F(t)$, $\Delta F_{\min} := \inf_{t \in \mathbb{R}} \Delta F(t)$, then*

$$d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q) = 2(\Delta F_{\max} - \Delta F_{\min}).$$

Combining the CDF perturbation behavior in Lem. 1 with the divergence characterization in Cor. 1, we obtain the following theorem.

Theorem 4. *The exchange rule does not decrease the $H\Delta H$ -divergence for the threshold hypothesis class*

$$d'_{H_{\text{thr}}\Delta H_{\text{thr}}}(P', Q') \geq d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q),$$

where $d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q)$ and $d'_{H_{\text{thr}}\Delta H_{\text{thr}}}(P', Q')$ denote the divergence before and after the swap, respectively.

Thms. 3 and 4 jointly demonstrate that the exchange mechanism increases mean separation and threshold-based distributional divergence, which provides a Bayes-optimal posterior guarantee for offline–online behavioral separability.

5.6. Algorithm Summary

We represent the learning agent as a collection of trainable components $\mathcal{C} = \{c_1, \dots, c_K\}$, which could include value functions, policies, or auxiliary networks depending on the underlying algorithm. The exchange mechanism operates on empirical statistics of alignment measure in Eq. 1, specifically the mean μ and standard deviation σ , estimated periodically from both offline and online buffers. Algorithm 1 summarizes the overall DARE procedure.

Algorithm 1 Dynamic Alignment for RELaxation (DARE)

- 1: **Initialize:** Offline components $\mathcal{C}$, behavior model π_b .
- 2: **Initialize:** Offline dataset $\mathcal{D}_{\text{off}}$, online buffer $\mathcal{D}_{\text{on}}$.
- 3: **for** each iteration **do**
- 4: Collect online samples and store them in $\mathcal{D}_{\text{on}}$.
- 5: Sample a mini-batch $\{b_{\text{off}}, b_{\text{on}}\}$ from $\mathcal{D}_{\text{off}}$ and $\mathcal{D}_{\text{on}}$.
- 6: Compute sample-wise alignment distance by Eq. 1.
- 7: Apply exchange rule to obtain exchange candidates.
- 8: Apply exchange operation to obtain $\{b'_{\text{off}}, b'_{\text{on}}\}$.
- 9: Update $\mathcal{C}$ using behavior-specific objectives.
- 10: **end for**

6. DARE Instantiations

The core of DARE is the exchange mechanism, which relies only on a per-sample alignment distance to a behavior model. Hence, it is compatible with a wide range of behavior models and fine-tuning objectives. We present representative instantiations for concreteness and empirical evaluation. Instantiation details are provided in Appendix C.

Behavior Model. We primarily adopt an energy-guided diffusion model (Lu et al., 2023b; Liu et al., 2024; Zhang et al., 2024b; Xu et al., 2025b) trained on the offline dataset to produce offline-consistent reference actions. As an alternative instantiation, we also consider the online policy actor for online alignment, which constitutes a general and interchangeable instantiation of the behavioral reference.

DARE in Cal-QL. We extend Cal-QL with DARE, namely the variant DARE-C. During online fine-tuning, the conservative constraint is enforced only on offline-like samples to preserve stability and removed for online-like samples to enable adaptation.

DARE in IQL. We extend IQL with DARE, denoted as DARE-I. DARE-I preserves the original IQL value targets and updates the policy using the PEX objective (Zhang et al., 2023) for offline-like samples. As for online-like samples, it replaces the value targets with TD-based maximum-Q estimates and adopts an entropy-regularized SAC objective (Haarnoja et al., 2018) for policy.

7. Experiments

We evaluate DARE and investigate the effectiveness, necessity, and robustness of the proposed exchange mechanism through the following key questions:

- **Q1** Does DARE improve offline-to-online performance across benchmarks? (see in the Sec. 7.2)
- **Q2** Does the exchange mechanism remain active and adaptive throughout training? (see in the Sec. 7.3)
- **Q3** Does a Bayes-optimal posterior play a necessary role in exchange decisions? (see in the Sec. 7.4)

Table 1. Performance after 0.2M online fine-tuning. Each result is averaged over the final 4 evaluations and 5 random seeds $\pm$ standard deviation. The “-C” and “-I” suffixes indicate the implementation based on Cal-QL and IQL, respectively. The highest scores are **bolded**.

Dataset	CQL Group				IQL Group			
	Base (CQL)	Cal-QL	EDIS-C	DARE-C	Base (IQL)	PEX	EDIS-I	DARE-I
HC-ME	96.3 $\pm$ 1.6	96.4 $\pm$ 0.9	95.1 $\pm$ 0.7	97.1$\pm$0.6	91.7 $\pm$ 2.3	89.2 $\pm$ 3.6	91.4 $\pm$ 3.9	93.6$\pm$1.2
H-ME	111.9 $\pm$ 0.9	111.9 $\pm$ 0.7	111.9 $\pm$ 1.7	111.9 $\pm$ 0.4	53.9 $\pm$ 39.0	90.2 $\pm$ 20.1	99.7 $\pm$ 13.5	104.5$\pm$7.8
W2D-ME	110.3 $\pm$ 0.5	110.4 $\pm$ 0.5	108.2 $\pm$ 7.1	112.3$\pm$1.3	111.8 $\pm$ 4.9	114.8$\pm$3.0	113.1 $\pm$ 0.9	114.9$\pm$2.1
HC-MR	50.9 $\pm$ 0.5	51.1 $\pm$ 1.1	56.4 $\pm$ 2.8	77.6$\pm$2.6	47.4 $\pm$ 1.0	53.3$\pm$1.2	45.7 $\pm$ 0.7	48.4 $\pm$ 0.7
H-MR	82.1 $\pm$ 33.2	93.0 $\pm$ 13.4	100.9 $\pm$ 5.9	102.7$\pm$1.1	87.0 $\pm$ 28.2	93.5 $\pm$ 13.7	93.7 $\pm$ 9.7	104.2$\pm$1.8
W2D-MR	86.9 $\pm$ 3.4	88.4 $\pm$ 4.6	108.9 $\pm$ 4.2	110.2$\pm$2.1	91.8 $\pm$ 6.2	92.0$\pm$6.4	89.2 $\pm$ 3.8	89.7 $\pm$ 3.2
HC-M	64.6 $\pm$ 2.6	66.9 $\pm$ 1.8	68.8 $\pm$ 1.8	77.2$\pm$3.4	57.8 $\pm$ 1.3	65.8 $\pm$ 2.9	49.3 $\pm$ 0.3	66.5$\pm$1.4
H-M	81.9 $\pm$ 8.1	86.6 $\pm$ 9.1	94.7 $\pm$ 7.2	100.0$\pm$0.8	77.5 $\pm$ 22.2	84.3 $\pm$ 20.1	58.9 $\pm$ 6.3	103.2$\pm$1.5
W2D-M	83.0 $\pm$ 0.7	83.4 $\pm$ 1.5	85.9 $\pm$ 1.5	89.2$\pm$3.2	85.8 $\pm$ 7.6	90.2 $\pm$ 13.1	86.4 $\pm$ 1.6	94.8$\pm$4.1
total (L)	767.9	788.1	830.8	878.2	704.7	773.3	727.4	819.8
AM-LD	53.8 $\pm$ 9.6	60.4 $\pm$ 7.7	70.1 $\pm$ 8.3	79.3$\pm$8.9	50.6 $\pm$ 7.3	53.5 $\pm$ 9.5	52.8 $\pm$ 2.7	53.9$\pm$5.3
AM-LP	50.2 $\pm$ 4.5	62.7 $\pm$ 9.5	68.2 $\pm$ 5.6	81.0$\pm$6.0	45.5 $\pm$ 7.6	50.8 $\pm$ 10.2	48.2 $\pm$ 6.5	52.6$\pm$4.2
AM-MD	85.8 $\pm$ 5.6	86.8 $\pm$ 4.7	93.4 $\pm$ 2.9	93.8$\pm$3.4	82.0 $\pm$ 6.0	82.8 $\pm$ 6.4	84.9$\pm$5.8	81.1 $\pm$ 3.6
AM-MP	86.2 $\pm$ 4.4	89.1 $\pm$ 5.4	94.4 $\pm$ 2.9	94.4 $\pm$ 5.2	80.4 $\pm$ 3.6	81.7 $\pm$ 4.7	78.1 $\pm$ 7.2	83.1$\pm$4.7
AM-UD	82.4 $\pm$ 7.4	90.1 $\pm$ 7.8	86.1 $\pm$ 3.7	91.3$\pm$6.1	31.6 $\pm$ 16.1	30.6 $\pm$ 13.4	34.9 $\pm$ 9.3	69.2$\pm$6.8
AM-U	93.2 $\pm$ 1.8	95.5 $\pm$ 1.3	95.2 $\pm$ 2.9	97.5$\pm$2.9	90.5 $\pm$ 3.5	92.9 $\pm$ 3.7	92.2 $\pm$ 2.3	93.3$\pm$2.7
total (AM)	451.6	484.6	507.4	537.3	380.6	392.3	391.1	433.2
total	1219.5	1272.7	1338.2	1415.4	1085.3	1165.6	1118.5	1253.0

Abbreviations: HC = halfcheetah, H = hopper, W2D = walker2d, L = locomotion, AM = antmaze; M = medium, ME = medium-expert, MR = medium-replay, LD = large-diverse, LP = large-play, MD = medium-diverse, MP = medium-play, UD = umaze-diverse, U = umaze. All environments use -v2 version.

- **Q4** Does DARE generalize across different behavior models and online objectives? (see in the Sec. 7.5)
- **Q5** Does violating behavior-consistent exchange degrade learning performance? (see in the Sec. 7.6)
- **Q6** Does removing the exchange mechanism impair effective learning? (see in the Sec. 7.7)

7.1. Baselines and Benchmarks

For baselines, we focus on methods that (i) can be *fully initialized* from the same offline-trained backbones and then fine-tuned online, and (ii) differ primarily in *constraint handling* rather than additional engineering choices (e.g., UTD, Q-ensemble). Accordingly, we group baselines by their offline initialization: in the **CQL group**, we compare DARE-C against CQL (Kumar et al., 2020), Cal-QL, and EDIS-C (Liu et al., 2024) (a Cal-QL variant); in the **IQL group**, we compare DARE-I against IQL, PEX (Zhang et al., 2023), and EDIS-I, the IQL counterpart of EDIS-C.

We evaluate DARE on three standard D4RL benchmarks (Fu et al., 2020): MuJoCo Locomotion, AntMaze Navigation, and Adroit Manipulation. For fair comparison, EDIS-C and DARE-C in the CQL group are initialized from the same Cal-QL models. Meanwhile, all methods in the IQL group are initialized from the same IQL models. Additional

implementation details, including the performance of the initial offline-trained models, are provided in **Appendix D**.

7.2. Overall Performance

The fine-tuning results are presented under both CQL group and IQL group in Tab. 1. Overall, DARE brings a significant improvement of around 11% total score of all datasets in both groups. Moreover, it consistently outperforms the baselines across different tasks, with highest scores on 13/15 tasks under the CQL group and on 12/15 tasks under the IQL group. These results demonstrate that the proposed exchange mechanism brings consistent benefits across diverse benchmarks in the O2O setting. **Appendix E** provides the fine-tuning learning curves, illustrating that DARE maintains stable improvements across locomotion tasks. In addition, **Appendix F** reports additional results on Kitchen tasks, different instantiations of DARE-I, and comparisons using the TD3+BC backbone (Fujimoto & Gu, 2021). Together, these results validate the effectiveness of DARE’s exchange mechanism across different experimental settings.

7.3. Visualization of Exchange Behavior

DARE applies the exchange mechanism to reassign samples between offline and online batches. As shown in Fig. 3,

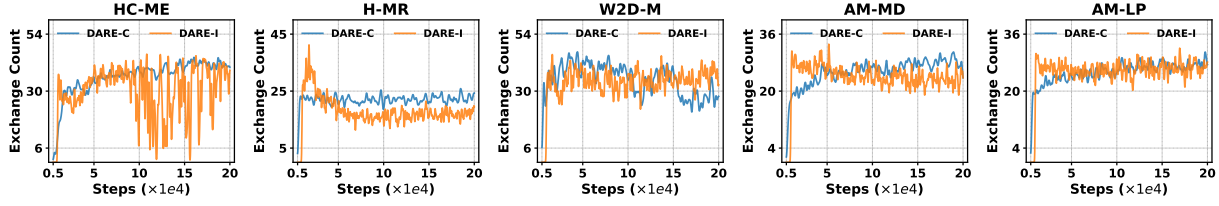
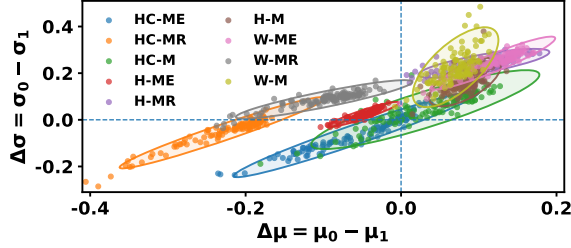


Figure 3. Adaptive exchange behavior of DARE-C and DARE-I during online training across different tasks.


 Figure 4. Estimated μ and σ gaps during $0.2M$ training steps, both computed using a diffusion model trained on the offline dataset.

the exchange pattern differs across environments, with the number of exchanges in most cases falling in the range of 20 to 50, which confirms that the proposed exchange mechanism actively performs sample reassignment during training. **Appendix G** further reports the exchange pattern for all environments. We further depict the evolution of the estimated Gaussian statistics used in the Bayes-optimal formulation, as shown in Fig. 4. A natural intuition suggests that the online buffer should induce a larger mean than the offline dataset. However, the empirical estimates do not consistently follow the expected mean-based ordering across environments or training stages. This observation indicates that the exchange decision cannot rely on a simple monotonic separation between offline and online statistics.

7.4. Effectiveness of Bayes-optimal Posterior

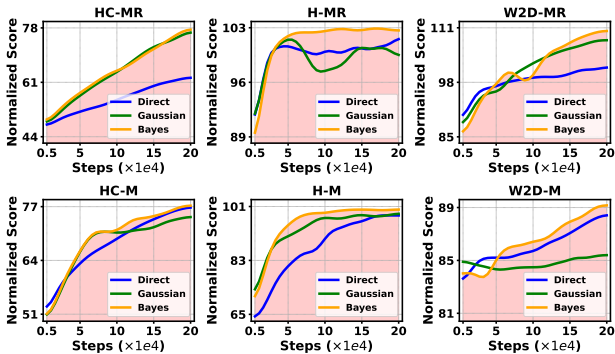


Figure 5. Comparison of posterior-induced exchange with monotonic Gaussian and direct alignment-distance on DARE-C.

We compare three exchange mechanisms based on the same alignment measure in Eq. 1: (i) our posterior-induced exchange mechanism using the posterior $\mathbb{P}(C|D)$; (ii) a monotonic Gaussian exchange based on fitted distributions; and (iii) direct alignment-distance exchange via sample-level sorting. Fig. 5 shows that the posterior-induced exchange achieves more stable performance than the two monotonic strategies. This result also aligns with Fig. 4, which do not exhibit a consistent monotonic ordering. As a consequence, monotonic exchange mechanisms become unreliable, whereas the Bayes-optimal formulation remains robust on the full posterior. Importantly, the exchange uses the Bayes-optimal posterior to relax constraints only for high-confidence samples, thereby preventing extrapolation-driven updates in uncertain regions.

7.5. Alternative Instantiations in DARE

We evaluate DARE on the Adroit manipulation suite to test the generality of the exchange mechanism under different online-objective instantiations. In these experiments, we keep the exchange mechanism fixed but only replace the online objective applied to online-like samples. As summarized in Tab. 2, although the absolute performance varies across objective instantiations, the exchange mechanism yields consistent improvements across Adroit tasks. A detailed analysis of these results is provided in **Appendix H**.

Table 2. Adroit results for different instantiations of DARE-I. w/o S (no SAC), w/o P (no PEX), w/o S&P (TD-only), with w/o E. and w/ E. indicating without and with exchange.

Adroit human-v0	DARE-I.w/o S		DARE-I.w/o P		DARE-I.w/o S&P	
	w/o E.	w/ E.	w/o E.	w/ E.	w/o E.	w/ E.
Door	-0.02	-0.15	-0.07	31.36	17.31	19.90
Hammer	0.18	0.26	-0.04	0.15	1.72	4.04
Pen	106.35	113.48	9.17	23.55	85.10	97.82
Relocate	0.15	0.27	-0.10	0.96	0.12	0.39
Total	106.67	113.90	8.96	56.02	104.25	122.15

We further consider an alternative instantiation of the behavior model in DARE. Instead of using a diffusion model trained on the offline dataset, we use the `online policy actor` as a consistency model for online-side behavior, reported in **Appendix I**. This instantiation shows slightly

lower performance than the diffusion-based offline behavior model. However, it avoids the computational cost of the reverse diffusion process and remains a practical option under limited computational budgets.

7.6. Reversed Behavior Assignment

Table 3. Results for reversed behavior assignment. Diffusion: offline diffusion as the behavior model, Policy: online policy actor as the behavior model, and DARE-I.C: curiosity-style DARE-I.

Locomotion	Diffusion		Policy	
	DARE-I.C	DARE-I	DARE-I.C	DARE-I
ME	306.2	313.0	294.7	311.7
MR	236.8	242.3	232.7	236.5
M	248.1	264.5	239.1	263.9
Total	791.1	819.8	766.5	812.1

DARE assumes that samples can be categorized by behavioral similarity and trained with behavior-specific objectives. To examine this assumption, we perform an experiment that reverses the assignment strategy: Identified online-like samples are reassigned to the offline batch, while offline-like samples are reassigned to the online batch. This configuration represents the opposite operation of DARE, corresponding to a curiosity-style assignment. We also impose an exchange limit constant $n = 32$, which matches the typical number of exchanges observed for DARE in Fig. 3. The corresponding results illustrated in Table 3, which supports behavioral consistency as the criterion for role assignment. More experimental results are provided in Appendix J.

7.7. Ablation Study

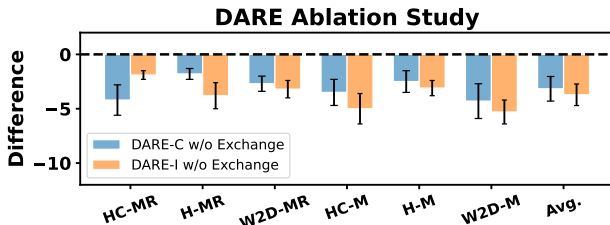


Figure 6. Ablation study of DARE.

Finally, we conduct an ablation study that removes the exchange mechanism, as shown in Fig. 6. The figure compares DARE-C and DARE-I with their corresponding w/o Exchange variants across multiple environments and reports the average performance difference. In all evaluated environments, both DARE-C and DARE-I show positive differences relative to their ablated counterparts, which reflects performance degradation after removal of the exchange operation. Additional ablation results are provided in Appendix K, including the choice of behavioral distance measure d in Eq. 1 and runtime comparisons.

8. Limitations

Although DARE improves offline-to-online transfer across different settings, its effectiveness still depends on the quality and coverage of the offline dataset. In DARE, the behavior model is first trained on offline data and then continuously updated during online fine-tuning. When the offline dataset has limited coverage or a substantial mismatch with the online trajectory, the behavior model may produce less reliable behavioral estimates, affecting the transition from offline regularization to online adaptation. In addition, our theoretical analysis focuses on the exchange rule rather than the end-to-end O2O learning process. A complete theory linking exchange decisions to value estimation and policy improvement remains an important direction for future work.

9. Conclusion

We address a fundamental limitation in O2O RL: distribution shift during fine-tuning induces out-of-support behaviors that existing approaches fail to handle at the sample level. To this end, we propose Dynamic Alignment for Relaxation (DARE), a distribution-aware framework that performs sample-level constraint relaxation based on behavioral consistency via a posterior-induced exchange mechanism. Theoretically, we demonstrate that the exchange mechanism follows from a Bayes-optimal posterior formulation and provably enhances offline-online behavioral separability. Importantly, DARE is algorithm-agnostic, admitting instantiations across diverse behavior-model choices and O2O algorithms. Empirically, DARE consistently improves performance over the corresponding baselines across all instantiated variants on standard benchmarks. These findings suggest that DARE provides a theoretically grounded foundation for bridging the offline-to-online gap through behavior-consistent, sample-level constraint adaptation.

Acknowledgements

The work of X. Zhang is partially supported by NSF under grants CCF-2312617, CCF-2431439, CNS-2453401, and CNS-2431553. The work of S. Chakraborty is partially supported by NSF under grant IIS-2143424 (NSF CAREER Award). We thank Hansong Zhou, Yukun Yuan, and the anonymous reviewers for their helpful feedback.

Impact Statement

This paper advances the study of offline-to-online RL by addressing the mismatch between data distributions and learning objectives during transfer. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

- Alonso, E., Jelley, A., Micheli, V., Kanervisto, A., Storkey, A. J., Pearce, T., and Fleuret, F. Diffusion for world modeling: Visual details matter in atari. *Advances in Neural Information Processing Systems*, 37:58757–58791, 2024.
- An, G., Moon, S., Kim, J.-H., and Song, H. O. Uncertainty-based offline reinforcement learning with diversified Q-ensemble. *Advances in neural information processing systems*, 34:7436–7447, 2021.
- Ball, P. J., Smith, L., Kostrikov, I., and Levine, S. Efficient online reinforcement learning with offline data. In *International Conference on Machine Learning*, pp. 1577–1594. PMLR, 2023.
- Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., and Vaughan, J. W. A theory of learning from different domains. *Machine learning*, 79(1):151–175, 2010.
- Chen, H., Lu, C., Ying, C., Su, H., and Zhu, J. Offline reinforcement learning via high-fidelity generative behavior modeling. In *International Conference on Learning Representations*, 2023.
- Feng, J., Feng, M., Song, H., Zhou, W., and Li, H. SUF: Stabilized unconstrained fine-tuning for offline-to-online reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 11961–11969, 2024.
- Fu, J., Kumar, A., Nachum, O., Tucker, G., and Levine, S. D4RL: Datasets for deep data-driven reinforcement learning. *arXiv preprint arXiv:2004.07219*, 2020.
- Fujimoto, S. and Gu, S. S. A minimalist approach to offline reinforcement learning. *Advances in neural information processing systems*, 34:20132–20145, 2021.
- Fujimoto, S., Meger, D., and Precup, D. Off-policy deep reinforcement learning without exploration. In *International conference on machine learning*, pp. 2052–2062. PMLR, 2019.
- Guo, R., Liu, L., Yang, R., Shen, J., Wu, G., Wang, J., and Li, B. Tackling heavy-tailed q-value bias in offline-to-online reinforcement learning with laplace-robust modeling. In *International Conference on Learning Representations*, 2026.
- Gupta, A., Kumar, V., Lynch, C., Levine, S., and Hausman, K. Relay policy learning: Solving long-horizon tasks via imitation and reinforcement learning. In *Conference on Robot Learning*, pp. 1025–1037. PMLR, 2020.
- Haarnoja, T., Zhou, A., Hartikainen, K., Tucker, G., Ha, S., Tan, J., Kumar, V., Zhu, H., Gupta, A., Abbeel, P., et al. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*, 2018.
- He, L., Ye, D., Tan, J., Wang, X., and Shen, L. Robust policy expansion for offline-to-online RL under diverse data corruption. *Advances in Neural Information Processing Systems*, 38:149262–149289, 2025.
- Hu, H., Yang, Y., Ye, J., Wu, C., Mai, Z., Hu, Y., Lv, T., Fan, C., Zhao, Q., and Zhang, C. Bayesian design principles for offline-to-online reinforcement learning. In *International Conference on Machine Learning*, pp. 19491–19515. PMLR, 2024.
- Janner, M., Du, Y., Tenenbaum, J., and Levine, S. Planning with diffusion for flexible behavior synthesis. In *International Conference on Machine Learning*, pp. 9902–9915. PMLR, 2022.
- Kostrikov, I., Nair, A., and Levine, S. Offline reinforcement learning with implicit Q-learning. In *International Conference on Learning Representations*, 2022.
- Kumar, A., Zhou, A., Tucker, G., and Levine, S. Conservative Q-learning for offline reinforcement learning. *Advances in neural information processing systems*, 33: 1179–1191, 2020.
- Lee, S., Seo, Y., Lee, K., Abbeel, P., and Shin, J. Offline-to-online reinforcement learning via balanced replay and pessimistic Q-ensemble. In *Conference on Robot Learning*, pp. 1702–1712. PMLR, 2022.
- Li, Q., Zhou, Z. P., and Levine, S. Reinforcement learning with action chunking. *Advances in Neural Information Processing Systems*, 38:55518–55553, 2025.
- Liu, X.-H., Liu, T.-S., Jiang, S., Chen, R., Zhang, Z., Chen, X., and Yu, Y. Energy-guided diffusion sampling for offline-to-online reinforcement learning. In *International Conference on Machine Learning*, pp. 31541–31565. PMLR, 2024.
- Lu, C., Ball, P., Teh, Y. W., and Parker-Holder, J. Synthetic experience replay. *Advances in Neural Information Processing Systems*, 36:46323–46344, 2023a.
- Lu, C., Chen, H., Chen, J., Su, H., Li, C., and Zhu, J. Contrastive energy prediction for exact energy-guided diffusion sampling in offline reinforcement learning. In *International Conference on Machine Learning*, pp. 22825–22855. PMLR, 2023b.
- Luo, Q.-W., Xie, M.-K., Wang, Y., and Huang, S.-J. Optimistic critic reconstruction and constrained fine-tuning for general offline-to-online RL. *Advances in Neural Information Processing Systems*, 37:108167–108207, 2024.

- Mao, Y., Wang, Q., Qu, Y., Jiang, Y., and Ji, X. Doubly mild generalization for offline reinforcement learning. *Advances in Neural Information Processing Systems*, 37: 51436–51473, 2024.
- Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., and Riedmiller, M. Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.
- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540): 529–533, 2015.
- Nair, A., Gupta, A., Dalal, M., and Levine, S. AWAC: Accelerating online reinforcement learning with offline datasets. *arXiv preprint arXiv:2006.09359*, 2020.
- Nakamoto, M., Zhai, S., Singh, A., Sobol Mark, M., Ma, Y., Finn, C., Kumar, A., and Levine, S. Cal-QL: Calibrated offline RL pre-training for efficient online fine-tuning. *Advances in Neural Information Processing Systems*, 36: 62244–62269, 2023.
- Rigter, M., Yamada, J., and Posner, I. World models via policy-guided trajectory diffusion. *Transactions on Machine Learning Research*, 2024(2), 2024.
- Shin, Y., Kim, J., Jung, W., Hong, S., Yoon, D., Jang, Y., Kim, G.-H., Chae, J., Sung, Y., Lee, K., et al. Online pre-training for offline-to-online reinforcement learning. In *International Conference on Machine Learning*, pp. 55122–55144. PMLR, 2025.
- Song, Y., Romero, A., Müller, M., Koltun, V., and Scaramuzza, D. Reaching the limit in autonomous racing: Optimal control versus reinforcement learning. *Science Robotics*, 8(82):eadg1462, 2023.
- Tarasov, D., Nikulin, A., Akimov, D., Kurenkov, V., and Kolesnikov, S. CORL: Research-oriented deep offline reinforcement learning library. *Advances in Neural Information Processing Systems*, 36:30997–31020, 2023.
- Towers, M., Kwiatkowski, A., Balis, J. U., Cola, G. D., Deleu, T., Goulão, M., Andreas, K., Krimmel, M., KG, A., Perez-Vicente, R. D. L., Terry, J. K., Pierré, A., Schulhoff, S. V., Tai, J. J., Tan, H., and Younis, O. G. Gymnasium: A standard interface for reinforcement learning environments. *Advances in Neural Information Processing Systems*, 38, 2025.
- Uehara, M., Zhao, Y., Hajiramezanali, E., Scalia, G., Eraslan, G., Lal, A., Levine, S., and Biancalani, T. Bridging model-based optimization and generative modeling via conservative fine-tuning of diffusion models. *Advances in Neural Information Processing Systems*, 37: 127511–127535, 2024.
- Van Hasselt, H., Guez, A., and Silver, D. Deep reinforcement learning with double Q-learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 30, 2016.
- Wang, S., Yang, Q., Gao, J., Lin, M., Chen, H., Wu, L., Jia, N., Song, S., and Huang, G. Train once, get a family: State-adaptive balances for offline-to-online reinforcement learning. *Advances in Neural Information Processing Systems*, 36:47081–47104, 2023.
- Wu, J., Wu, H., Qiu, Z., Wang, J., and Long, M. Supported policy optimization for offline reinforcement learning. *Advances in Neural Information Processing Systems*, 35: 31278–31291, 2022.
- Xie, T., Jiang, N., Wang, H., Xiong, C., and Bai, Y. Policy finetuning: Bridging sample-efficient offline and online reinforcement learning. *Advances in neural information processing systems*, 34:27395–27407, 2021.
- Xu, H., Mao, L., Jin, H., Zhang, W., Zhan, X., and Zhang, A. Uni-RL: Unifying online and offline RL via implicit value regularization. *Advances in Neural Information Processing Systems*, 38:133162–133187, 2025a.
- Xu, M., Geffner, T., Kreis, K., Nie, W., Xu, Y., Leskovec, J., Ermon, S., and Vahdat, A. Energy-based diffusion language models for text generation. In *International Conference on Learning Representations*, pp. 33769–33789, 2025b.
- Yang, R., Bai, C., Ma, X., Wang, Z., Zhang, C., and Han, L. RORL: Robust offline reinforcement learning via conservative smoothing. *Advances in neural information processing systems*, 35:23851–23866, 2022.
- Yu, Z. and Zhang, X. Actor-critic alignment for offline-to-online reinforcement learning. In *International Conference on Machine Learning*, pp. 40452–40474. PMLR, 2023.
- Zhang, H., Xu, W., and Yu, H. Policy expansion for bridging offline-to-online reinforcement learning. In *International Conference on Learning Representations*, 2023.
- Zhang, R. and Zanette, A. Policy finetuning in reinforcement learning via design of experiments using offline data. *Advances in Neural Information Processing Systems*, 36: 59953–59995, 2023.
- Zhang, Y., Liu, J., Li, C., Niu, Y., Yang, Y., Liu, Y., and Ouyang, W. A perspective of Q-value estimation on offline-to-online reinforcement learning. In *Proceedings*

of the AAAI conference on artificial intelligence, volume 38, pp. 16908–16916, 2024a.

Zhang, Y., Yu, P., and Wu, Y. N. Object-conditioned energy-based attention map alignment in text-to-image diffusion models. In *European Conference on Computer Vision*, pp. 55–71. Springer, 2024b.

Zheng, H., Luo, X., Wei, P., Song, X., Li, D., and Jiang, J. Adaptive policy learning for offline-to-online reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pp. 11372–11380, 2023.

Zheng, H., Shen, L., Luo, Y., Ye, D., Xu, S., Du, B., Shen, J., and Tao, D. Value-guided decision transformer: A unified reinforcement learning framework for online and offline settings. *Advances in Neural Information Processing Systems*, 38:71655–71684, 2025.

Appendix Guide

We provide the supporting details for the main paper, organized as follows:

A	Proofs	14
Thm. 1	Sufficiency of d for Bayes-optimal Posterior Inference	14
Thm. 2	Stationary Point Location for the Quadratic-logistic Posterior	15
Prop. 1	Case-wise Form of the Sign-based Exchange Rule	16
Thm. 3	Increase of the Absolute Mean Separation	16
Lemm. 1	Piecewise Characterization of Post-exchange CDFs	17
Cor. 1	Closed Form of $H_{\text{thr}}\Delta H_{\text{thr}}$ via CDFs	18
Thm. 4	Non-decreasing Threshold Separation under Exchange	19
B	Analysis of the Gaussian Assumption	20
C	DARE Instantiation Details	21
D	Experimental Details	22
E	Comparisons of Online Fine-tuning Processes	25
F	Additional Results for Main Table	26
G	Visualization of Exchange Behavior	27
H	Further Results and Analysis on Adroit	28
I	Alternative Behavior Model Instantiations	29
J	Additional Results for Reversed Behavior Assignment	31
K	Additional Ablation Study	32

A. Proofs

Definition 1. Let (s, a) denote a state–action pair taking values in space $\mathcal{S} \times \mathcal{A}$, and let $\pi_b(\cdot \mid s)$ denote a reference behavior model. Define a one-dimensional behavioral deviation statistic

$$d(s, a) := \text{dist}(a, a_b(s)), \quad \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R},$$

where $\text{dist}(\cdot, \cdot)$ is a fixed deviation measure and $a_b(s)$ denotes the reference action by π_b .

Theorem 1. Suppose the class-conditional densities admit a likelihood–ratio representation through d ; that is, there exists a measurable function $r : \mathbb{R} \rightarrow (0, \infty)$ such that

$$\frac{p(X = x \mid C = 0)}{p(X = x \mid C = 1)} = r(d(s, a)).$$

It follows that d is sufficient for Bayes–optimal posterior inference between the two classes, i.e. there exists a measurable function $g : \mathbb{R} \rightarrow [0, 1]$ such that

$$\mathbb{P}(C = 0 \mid X = x) = g(d(s, a)).$$

Proof. Fix any state-conditioned action variable x and by Bayes’ rule, the posterior probability can be written as

$$\mathbb{P}(C = 0 \mid X = x) = \frac{\mathbb{P}(C = 0) p_0(x)}{\mathbb{P}(C = 0) p_0(x) + \mathbb{P}(C = 1) p_1(x)}.$$

Dividing the numerator and denominator by $p_1(x)$ yields

$$\mathbb{P}(C = 0 \mid X = x) = \frac{\mathbb{P}(C = 0) \frac{p_0(x)}{p_1(x)}}{\mathbb{P}(C = 0) \frac{p_0(x)}{p_1(x)} + \mathbb{P}(C = 1)} = \frac{\mathbb{P}(C = 0) r(d(s, a))}{\mathbb{P}(C = 0) r(d(s, a)) + \mathbb{P}(C = 1)}.$$

Since the right-hand side is a function of $r(d(s, a))$ alone, there exists a measurable function $g : \mathbb{R} \rightarrow (0, 1)$ such that

$$\mathbb{P}(C = 0 \mid X = x) = g(d(s, a)).$$

□

Property 1. Let $D : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ be a real-valued random variable and let $C \in \{0, 1\}$ denote a binary class label. We say that the posterior distribution of C given D is second-order Gaussian-determined if there exist parameters $\mu_0, \mu_1 \in \mathbb{R}$ and $\sigma_0^2, \sigma_1^2 > 0$ such that

$$D \mid C = c \sim \mathcal{N}(\mu_c, \sigma_c^2), \quad c \in \{0, 1\}.$$

Under this model, the class-conditional densities of D are given by

$$p(D = d \mid C = 0) = \frac{1}{\sqrt{2\pi} \sigma_0} \exp\left(-\frac{(d - \mu_0)^2}{2\sigma_0^2}\right), \quad p(D = d \mid C = 1) = \frac{1}{\sqrt{2\pi} \sigma_1} \exp\left(-\frac{(d - \mu_1)^2}{2\sigma_1^2}\right).$$

Property 2. Let $\rho_c := \mathbb{P}(C = c)$ with $\rho_0, \rho_1 > 0$ and $\rho_0 + \rho_1 = 1$. Under this model, the posterior probability $\mathbb{P}(C = 0 \mid D = d)$. Then, for any $d \in \mathbb{R}$, the posterior probability is

$$\mathbb{P}(C = 0 \mid D = d) = \frac{\rho_0 \sigma_1 \exp\left(-\frac{(d - \mu_0)^2}{2\sigma_0^2}\right)}{\rho_0 \sigma_1 \exp\left(-\frac{(d - \mu_0)^2}{2\sigma_0^2}\right) + \rho_1 \sigma_0 \exp\left(-\frac{(d - \mu_1)^2}{2\sigma_1^2}\right)} = \frac{1}{1 + \frac{\rho_1 \sigma_0}{\rho_0 \sigma_1} \exp\left(-\frac{(d - \mu_1)^2}{2\sigma_1^2} + \frac{(d - \mu_0)^2}{2\sigma_0^2}\right)}$$

Property 3. Let $\alpha > 0$ and define

$$f(x) := \frac{1}{1 + \alpha \exp(-x)}.$$

Then f is strictly increasing in x , with derivative

$$f'(x) = \frac{\alpha \exp(-x)}{(1 + \alpha \exp(-x))^2} > 0.$$

Under Property 2, define

$$x : \ell(d) = \frac{(d - \mu_1)^2}{2\sigma_1^2} - \frac{(d - \mu_0)^2}{2\sigma_0^2}, \quad \alpha := \frac{\rho_1 \sigma_0}{\rho_0 \sigma_1}.$$

Then the posterior can be written as

$$\mathbb{P}(C = 0 \mid D = d) = f(\ell(d)).$$

Moreover, $\ell(d)$ is a quadratic function of d :

$$\ell(d) = \ell_1 d^2 + \ell_2 d + \ell_3,$$

where the quadratic coefficient is

$$\ell_1 = \frac{1}{2} \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2} \right), \quad \ell_2 = -\frac{\mu_1}{\sigma_1^2} + \frac{\mu_0}{\sigma_0^2}, \quad \ell_3 = \frac{\mu_1^2}{2\sigma_1^2} - \frac{\mu_0^2}{2\sigma_0^2}.$$

Consequently,

$$x'(d) = 2\ell_1 d + \ell_2,$$

and when $\alpha \neq 0$, the unique stationary point (extremum) of $x(d)$ is

$$d^* = -\frac{\ell_2}{2\ell_1} = \frac{\mu_1/\sigma_1^2 - \mu_0/\sigma_0^2}{1/\sigma_1^2 - 1/\sigma_0^2}.$$

Theorem 2. The stationary point d^* in Property 3 satisfies $d^* \notin (\mu_0, \mu_1)$.

Proof. From Property 3,

$$d^* = \frac{\mu_1 \sigma_0^2 - \mu_0 \sigma_1^2}{\sigma_0^2 - \sigma_1^2} = \mu_0 + (\mu_1 - \mu_0) \frac{\sigma_0^2}{\sigma_0^2 - \sigma_1^2}.$$

Let $r := \frac{\sigma_0^2}{\sigma_0^2 - \sigma_1^2}$. Since $\sigma_0^2 \neq \sigma_1^2$, either $r > 1$ (when $\sigma_0^2 > \sigma_1^2$) or $r < 0$ (when $\sigma_0^2 < \sigma_1^2$). We consider four cases:

Case 1: $\mu_0 < \mu_1$ and $\sigma_0^2 > \sigma_1^2$. Then $\mu_1 - \mu_0 > 0$ and $r > 1$, hence

$$d^* = \mu_0 + (\mu_1 - \mu_0)r > \mu_0 + (\mu_1 - \mu_0) = \mu_1.$$

Case 2: $\mu_0 < \mu_1$ and $\sigma_0^2 < \sigma_1^2$. Then $\mu_1 - \mu_0 > 0$ and $r < 0$, hence

$$d^* = \mu_0 + (\mu_1 - \mu_0)r < \mu_0.$$

Case 3: $\mu_1 < \mu_0$ and $\sigma_0^2 > \sigma_1^2$. Then $\mu_1 - \mu_0 < 0$ and $r > 1$, hence

$$d^* = \mu_0 + (\mu_1 - \mu_0)r < \mu_0 + (\mu_1 - \mu_0) = \mu_1.$$

Case 4: $\mu_1 < \mu_0$ and $\sigma_0^2 < \sigma_1^2$. Then $\mu_1 - \mu_0 < 0$ and $r < 0$, hence

$$d^* = \mu_0 + (\mu_1 - \mu_0)r > \mu_0.$$

In all cases, d^* lies outside the open interval with endpoints μ_0 and μ_1 , i.e., $d^* \notin (\min\{\mu_0, \mu_1\}, \max\{\mu_0, \mu_1\})$. In particular, under $\mu_0 < \mu_1$, we have $d^* \notin (\mu_0, \mu_1)$. $\square$

Definition 2. Let $D : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ be a real-valued random variable and let $C \in \{0, 1\}$ denote a binary class label. Denote the class-wise means and standard deviations by $\mu_0, \mu_1, \sigma_0, \sigma_1$. Define the mean-direction and dispersion-direction signs as

$$\delta_\mu := \text{sign}(\mu_0 - \mu_1), \quad \delta_\sigma := \text{sign}(\sigma_0 - \sigma_1),$$

and the thresholds

$$\tau := \frac{\mu_0 + \mu_1}{2}, \quad \tau_{\text{side}} := 2d^* - \tau,$$

where d^* is stationary point.

An element with projection value d belonging to class $C = 0$ is selected for exchange to class $C = 1$ if and only if

$$\delta_\mu \delta_\sigma (d - \tau_{\text{side}}) > 0 \quad \text{and} \quad \delta_\mu (d - \tau) < 0.$$

An element with projection value d belonging to class $C = 1$ is selected for exchange to class $C = 0$ if and only if

$$\delta_\mu \delta_\sigma (d - \tau_{\text{side}}) > 0 \quad \text{and} \quad \delta_\mu (d - \tau) > 0.$$

Proposition 1. *The sign-based exchange rule in Definition 2 is equivalent to the following four case-by-case conditions:*

Case 1: If $\mu_0 > \mu_1$ and $\sigma_0 > \sigma_1$ (i.e., $\delta_\mu = +1$, $\delta_\sigma = +1$), then

$$(0 \rightarrow 1) : \tau_{\text{side}} < d < \tau, \quad (1 \rightarrow 0) : d > \max\{\tau, \tau_{\text{side}}\}.$$

Case 2: If $\mu_0 > \mu_1$ and $\sigma_0 < \sigma_1$ (i.e., $\delta_\mu = +1$, $\delta_\sigma = -1$), then

$$(0 \rightarrow 1) : d < \min\{\tau, \tau_{\text{side}}\}, \quad (1 \rightarrow 0) : \tau < d < \tau_{\text{side}}.$$

Case 3: If $\mu_0 < \mu_1$ and $\sigma_0 > \sigma_1$ (i.e., $\delta_\mu = -1$, $\delta_\sigma = +1$), then

$$(0 \rightarrow 1) : \tau < d < \tau_{\text{side}}, \quad (1 \rightarrow 0) : d < \min\{\tau, \tau_{\text{side}}\}.$$

Case 4: If $\mu_0 < \mu_1$ and $\sigma_0 < \sigma_1$ (i.e., $\delta_\mu = -1$, $\delta_\sigma = -1$), then

$$(0 \rightarrow 1) : d > \max\{\tau, \tau_{\text{side}}\}, \quad (1 \rightarrow 0) : \tau_{\text{side}} < d < \tau.$$

Proof. For class $0 \rightarrow 1$, the rule requires simultaneously $\delta_\mu \delta_\sigma (d - \tau_{\text{side}}) > 0$ and $\delta_\mu (d - \tau) < 0$. If $\delta_\mu = +1$, the second inequality is $d < \tau$; if $\delta_\mu = -1$, it is $d > \tau$. Likewise, if $\delta_\mu \delta_\sigma = +1$, the first inequality is $d > \tau_{\text{side}}$, whereas if $\delta_\mu \delta_\sigma = -1$, it becomes $d < \tau_{\text{side}}$. Intersecting these two half-spaces yields exactly the interval conditions listed for $(0 \rightarrow 1)$ in each case.

The class $1 \rightarrow 0$ rule is identical except that the second inequality flips to $\delta_\mu (d - \tau) > 0$, which becomes $d > \tau$ when $\delta_\mu = +1$ and $d < \tau$ when $\delta_\mu = -1$. Combining with the same first inequality $\delta_\mu \delta_\sigma (d - \tau_{\text{side}}) > 0$ yields the listed conditions for $(1 \rightarrow 0)$ in each case. $\square$

Theorem 3. *Let $\mathcal{B}_0, \mathcal{B}_1$ be two finite sets of projected values with empirical means μ_0, μ_1 and sizes n_0, n_1 . Define*

$$\Delta := \mu_0 - \mu_1, \quad \delta_\mu := \text{sign}(\Delta), \quad \tau := \frac{\mu_0 + \mu_1}{2}.$$

Select $x \in \mathcal{B}_0$ and $y \in \mathcal{B}_1$ such that

$$\delta_\mu (x - \tau) < 0 \quad \text{and} \quad \delta_\mu (y - \tau) > 0,$$

and swap x and y between the two sets. Let μ'_0, μ'_1 denote the empirical means after the exchange and define $\Delta' := \mu'_0 - \mu'_1$. Each exchange strictly increases the absolute mean gap:

$$|\Delta'| > |\Delta|.$$

Proof. After swapping $x \in \mathcal{B}_0$ with $y \in \mathcal{B}_1$, the batch sizes remain n_0, n_1 and the new means satisfy

$$\mu'_0 = \mu_0 + \frac{y - x}{n_0}, \quad \mu'_1 = \mu_1 + \frac{x - y}{n_1}.$$

Hence, the new mean gap is

$$\Delta' = \mu'_0 - \mu'_1 = (\mu_0 - \mu_1) + \left(\frac{1}{n_0} + \frac{1}{n_1} \right) (y - x) = \Delta + \kappa (y - x),$$

where $\kappa := \frac{1}{n_0} + \frac{1}{n_1} > 0$.

It remains to show that $\delta_\mu(y - x) > 0$. From the side conditions,

$$\delta_\mu(x - \tau) < 0 \iff \delta_\mu x < \delta_\mu \tau, \quad \delta_\mu(y - \tau) > 0 \iff \delta_\mu y > \delta_\mu \tau.$$

Combining them yields

$$\delta_\mu y > \delta_\mu \tau > \delta_\mu x \implies \delta_\mu(y - x) > 0.$$

Therefore,

$$\delta_\mu \Delta' = \delta_\mu \Delta + \kappa \delta_\mu(y - x) = |\Delta| + \kappa |y - x| > |\Delta|.$$

Since $\delta_\mu \Delta' > 0$, we also have $\text{sign}(\Delta') = \delta_\mu = \text{sign}(\Delta)$.

Finally,

$$|\Delta'| = \delta_\mu \Delta' = |\Delta| + \kappa |y - x| \implies |\Delta'| > |\Delta|.$$

This proves the claim. $\square$

Lemma 1. *Consider the empirical CDF*

$$F_c(t) = \frac{1}{n_c} \sum_{i=1}^{n_c} \mathbf{1}\{d_i^{(c)} \leq t\}, \quad c \in \{0, 1\}.$$

Suppose a single element with value x in class c is swapped with an element with value y from the opposite class. Let F'_c denote the resulting empirical CDF after the swap. Then, for every $t \in \mathbb{R}$,

$$F'_c(t) = F_c(t) - (-1)^c \delta_\mu \frac{1}{n_c} \mathbf{1}\{\delta_\mu x \leq \delta_\mu t < \delta_\mu y\}.$$

Proof. By construction, all projected values remain unchanged except for the replacement of x by y in class 0 and the replacement of y by x in class 1. Hence, for any t ,

$$F'_0(t) - F_0(t) = \frac{1}{n_0} \left(\mathbf{1}\{y \leq t\} - \mathbf{1}\{x \leq t\} \right), \quad F'_1(t) - F_1(t) = \frac{1}{n_1} \left(\mathbf{1}\{x \leq t\} - \mathbf{1}\{y \leq t\} \right).$$

It remains to simplify these indicator differences under the selection conditions. Since $\delta_\mu(x - \tau) < 0$ and $\delta_\mu(y - \tau) > 0$, we have

$$\delta_\mu x < \delta_\mu \tau < \delta_\mu y \implies \delta_\mu(y - x) > 0.$$

Consider the event

$$I = \{ \delta_\mu y \leq \delta_\mu t < \delta_\mu x \} = \begin{cases} \{t : y \leq t < x\}, & \text{if } \delta_\mu = +1, \\ \{t : x \leq t < y\}, & \text{if } \delta_\mu = -1. \end{cases}$$

For $t \notin I$, either t is on the same side of both x and y (so $\mathbf{1}\{y \leq t\} = \mathbf{1}\{x \leq t\}$), or t exceeds both (so again the two indicators coincide). Thus, $F'_0(t) = F_0(t)$ and $F'_1(t) = F_1(t)$ for $t \notin I$.

For $t \in I$, exactly one of $\{x \leq t\}$ and $\{y \leq t\}$ holds, and since $\delta_\mu(y - x) > 0$, we have

$$\mathbf{1}\{y \leq t\} - \mathbf{1}\{x \leq t\} = -\delta_\mu, \quad \mathbf{1}\{x \leq t\} - \mathbf{1}\{y \leq t\} = \delta_\mu.$$

Substituting into the CDF differences yields

$$F'_0(t) = F_0(t) - \delta_\mu \frac{1}{n_0}, \quad F'_1(t) = F_1(t) + \delta_\mu \frac{1}{n_1}, \quad \forall t \in I,$$

which is precisely

$$F'_0(t) = F_0(t) - \delta_\mu \frac{1}{n_0} \mathbf{1}\{t \in I\}, \quad F'_1(t) = F_1(t) + \delta_\mu \frac{1}{n_1} \mathbf{1}\{t \in I\}.$$

This completes the proof. $\square$

Definition 3. We consider the $H\Delta H$ -divergence introduced by Ben-David et al. (2010), instantiated with the one-dimensional threshold hypothesis class

$$H_{\text{thr}} = \{h_t(z) = \mathbf{1}\{z \leq t\} : t \in \mathbb{R}\}.$$

For two distributions P and Q over $Z \in \mathbb{R}$, the $H_{\text{thr}}\Delta H_{\text{thr}}$ -divergence is defined as

$$d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q) := 2 \sup_{h, h' \in H_{\text{thr}}} |P(h(Z) \neq h'(Z)) - Q(h(Z) \neq h'(Z))|.$$

Corollary 1. Let F_0 and F_1 denote the cumulative distribution functions (CDFs) of P and Q , respectively. Define

$$\Delta F(t) := F_0(t) - F_1(t), \quad \Delta F_{\max} := \sup_{t \in \mathbb{R}} \Delta F(t), \quad \Delta F_{\min} := \inf_{t \in \mathbb{R}} \Delta F(t).$$

Then

$$d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q) = 2(\Delta F_{\max} - \Delta F_{\min}).$$

Proof. For $a < b$, note that for the threshold functions $h_a(z) = \mathbf{1}\{z \leq a\}$ and $h_b(z) = \mathbf{1}\{z \leq b\}$, their symmetric difference satisfies

$$h_a(z) \oplus h_b(z) = \mathbf{1}\{a < z \leq b\}.$$

Conversely, every interval indicator $\mathbf{1}\{a < z \leq b\}$ can be written as $h_a \oplus h_b$ for some $h_a, h_b \in H_{\text{thr}}$. Therefore, by definition of the $H\Delta H$ -divergence with base class H_{thr} ,

$$d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q) = 2 \sup_{a < b} |P(a < Z \leq b) - Q(a < Z \leq b)|.$$

Using the identity $\{a < Z \leq b\} = \{Z \leq b\} \cap \{Z \leq a\}^c$, we obtain

$$P(a < Z \leq b) = F_0(b) - F_0(a), \quad Q(a < Z \leq b) = F_1(b) - F_1(a),$$

and hence

$$d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q) = 2 \sup_{a < b} |\Delta F(b) - \Delta F(a)|.$$

Since $\Delta F_{\min} \leq \Delta F(t) \leq \Delta F_{\max}$ for all $t \in \mathbb{R}$, we have

$$|\Delta F(b) - \Delta F(a)| \leq \Delta F_{\max} - \Delta F_{\min}, \quad \forall a < b.$$

Conversely, choose sequences $\{t_n^{\max}\}$ and $\{t_n^{\min}\}$ such that $\Delta F(t_n^{\max}) \rightarrow \Delta F_{\max}$ and $\Delta F(t_n^{\min}) \rightarrow \Delta F_{\min}$. For each n , define

$$a_n := \min\{t_n^{\max}, t_n^{\min}\}, \quad b_n := \max\{t_n^{\max}, t_n^{\min}\},$$

so that $a_n < b_n$. Then

$$|\Delta F(b_n) - \Delta F(a_n)| = |\Delta F(t_n^{\max}) - \Delta F(t_n^{\min})| \rightarrow \Delta F_{\max} - \Delta F_{\min}.$$

Hence,

$$\sup_{a < b} |\Delta F(b) - \Delta F(a)| \geq \Delta F_{\max} - \Delta F_{\min}.$$

Combining the upper and lower bounds yields

$$d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q) = 2(\Delta F_{\max} - \Delta F_{\min}),$$

which completes the proof. $\square$

Theorem 4. Let $\mathcal{B}_0, \mathcal{B}_1$ be two finite sets of projected values and select $x \in \mathcal{B}_0$ and $y \in \mathcal{B}_1$ such that $\delta_\mu(x - \tau) < 0$ and $\delta_\mu(y - \tau) > 0$ for exchange between the two sets. Then any exchange rule does not decrease the $H\Delta H$ -divergence for the threshold hypothesis class:

$$d'_{H_{\text{thr}}\Delta H_{\text{thr}}}(P', Q') \geq d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q),$$

where $d_{H_{\text{thr}}\Delta H_{\text{thr}}}$ and $d'_{H_{\text{thr}}\Delta H_{\text{thr}}}$ denote the divergence before and after the swap, respectively.

Proof. By Corollary 1,

$$d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q) = 2 \left(\sup_{t \in \mathbb{R}} \Delta F(t) - \inf_{t \in \mathbb{R}} \Delta F(t) \right), \quad d'_{H_{\text{thr}}\Delta H_{\text{thr}}}(P', Q') = 2 \left(\sup_{t \in \mathbb{R}} \Delta F'(t) - \inf_{t \in \mathbb{R}} \Delta F'(t) \right).$$

Hence it suffices to show that the range of ΔF does not shrink:

$$\left(\sup \Delta F' - \inf \Delta F' \right) \geq \left(\sup \Delta F - \inf \Delta F \right).$$

By Lemma 1,

$$F'_0(t) = F_0(t) - \delta_\mu \frac{1}{n_0} \mathbf{1}\{t \in I\}, \quad F'_1(t) = F_1(t) + \delta_\mu \frac{1}{n_1} \mathbf{1}\{t \in I\}, \quad \text{where } I := \{t \in \mathbb{R} : \delta_\mu x \leq \delta_\mu t < \delta_\mu y\}.$$

Therefore

$$\Delta F'(t) = \Delta F(t) - \delta_\mu \kappa \mathbf{1}\{t \in I\}, \quad \forall t \in \mathbb{R}. \quad (9)$$

The swap leaves ΔF unchanged outside I , and shifts it by a constant $-\delta_\mu \kappa$ on I .

Let $M := \sup_t \Delta F(t)$ and $m := \inf_t \Delta F(t)$, and similarly $M' := \sup_t \Delta F'(t)$ and $m' := \inf_t \Delta F'(t)$. Write the domain decomposition $\mathbb{R} = I \cup I^c$. From (9),

$$\sup_{t \in I^c} \Delta F'(t) = \sup_{t \in I^c} \Delta F(t), \quad \inf_{t \in I^c} \Delta F'(t) = \inf_{t \in I^c} \Delta F(t),$$

and

$$\begin{aligned} \sup_{t \in I} \Delta F'(t) &= \sup_{t \in I} (\Delta F(t) - \delta_\mu \kappa) = \sup_{t \in I} \Delta F(t) - \delta_\mu \kappa, \\ \inf_{t \in I} \Delta F'(t) &= \inf_{t \in I} (\Delta F(t) - \delta_\mu \kappa) = \inf_{t \in I} \Delta F(t) - \delta_\mu \kappa. \end{aligned}$$

Hence

$$\begin{aligned} M' &= \max \left\{ \sup_{t \in I^c} \Delta F(t), \sup_{t \in I} \Delta F(t) - \delta_\mu \kappa \right\}, \\ m' &= \min \left\{ \inf_{t \in I^c} \Delta F(t), \inf_{t \in I} \Delta F(t) - \delta_\mu \kappa \right\}. \end{aligned}$$

Now consider two cases.

Case A: $\delta_\mu = +1$. Then ΔF is shifted *downward* by κ on I . Therefore $M' \geq \sup_{t \in I^c} \Delta F(t)$ and $m' \leq \inf_{t \in I} \Delta F(t) - \kappa \leq m - \kappa$ whenever the global infimum is attained on I ; otherwise $m' = m$. In either situation, the maximum cannot increase, and the minimum cannot increase either; indeed the only possible change is that the function becomes *no larger* on I . Consequently,

$$M' - m' \geq M - m,$$

because either the range is unchanged (if both extrema lie in I^c , or both lie in I), or the minimum decreases (if the infimum is attained on I), which enlarges the range.

Case B: $\delta_\mu = -1$. Then ΔF is shifted *upward* by κ on I . A symmetric argument shows that m' cannot decrease while M' cannot decrease; the only possible change is that the function becomes *no smaller* on I . Hence again

$$M' - m' \geq M - m.$$

Combining the range inequality with Corollary 1 yields

$$d'_{H_{\text{thr}}\Delta H_{\text{thr}}}(P', Q') = 2(M' - m') \geq 2(M - m) = d_{H_{\text{thr}}\Delta H_{\text{thr}}}(P, Q).$$

This proves the claim. $\square$

B. Analysis of the Gaussian Assumption

The Gaussian assumption in our analysis is used to obtain a tractable closed-form characterization of the quadratic term in the likelihood-ratio expression. To examine whether the key conclusion depends critically on this specific distributional choice, we consider an alternative Gamma-family model for the nonnegative behavior-distance variable d . We show that the stationary point induced by the Gamma likelihood ratio only introduces a higher-order perturbation relative to the Gaussian case. More importantly, the Gamma distribution preserves the conclusion in Thm. 2.

Stationary point of the Gamma density ratio. Consider two Gamma distributions parameterized by shape α and rate β

$$f_1(d; \alpha_1, \beta_1) = \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)} d^{\alpha_1-1} e^{-\beta_1 d}, \quad f_2(d; \alpha_2, \beta_2) = \frac{\beta_2^{\alpha_2}}{\Gamma(\alpha_2)} d^{\alpha_2-1} e^{-\beta_2 d}, \quad d > 0.$$

Their likelihood ratio can be formatted as

$$\frac{f_1(d)}{f_2(d)} = \frac{\Gamma(\alpha_2)\beta_1^{\alpha_1}}{\Gamma(\alpha_1)\beta_2^{\alpha_2}} d^{\alpha_1-\alpha_2} e^{-(\beta_1-\beta_2)d} = C_\Gamma d^{\alpha_1-\alpha_2} e^{-(\beta_1-\beta_2)d},$$

where $C_\Gamma = \frac{\Gamma(\alpha_2)\beta_1^{\alpha_1}}{\Gamma(\alpha_1)\beta_2^{\alpha_2}}$ is independent of d .

Taking the logarithm and differentiating with respect to d yields the stationary point

$$d_{\text{Gamma}}^* = \frac{\alpha_1 - \alpha_2}{\beta_1 - \beta_2} = \frac{\mu_1^2 \sigma_2^2 - \mu_2^2 \sigma_1^2}{\mu_1 \sigma_2^2 - \mu_2 \sigma_1^2}, \text{ by } \alpha = \frac{\mu^2}{\sigma^2} \text{ and } \beta = \frac{\mu}{\sigma^2}.$$

Limited perturbation from Gamma modeling. Compared to its Gaussian counterpart in Eq. 5, the difference is

$$\delta_p = d_{\text{Gamma}}^* - d_{\text{Gaussian}}^* = - \frac{(\mu_1 - \mu_2)^2 \sigma_1^2 \sigma_2^2}{(\mu_1 \sigma_2^2 - \mu_2 \sigma_1^2)(\sigma_2^2 - \sigma_1^2)}.$$

Under the non-degenerate condition that the denominator is bounded away from zero, this discrepancy is quadratic in the mean separation $(\mu_1 - \mu_2)$. In Tab. 4, we further show that adopting a Gamma-family model only introduces a limited perturbation, which supports the use of the Gaussian model as a tractable approximation in our framework.

Table 4. The reported values are averaged over the last 50 checkpoints, recorded every 1000 training steps.

Dataset	d_{Gamma}^*	d_{Gaussian}^*	$ \delta_p $
H-ME	0.2849	0.3315	0.0465
H-MR	0.2028	0.2427	0.0399
H-M	0.2899	0.3203	0.0303
W2D-ME	0.1537	0.1810	0.0273
W2D-MR	0.5786	0.6243	0.0457
W2D-M	0.3583	0.3919	0.0336

Preserved outside-interval property under Gamma modeling. Under the Gamma-family model, we have

$$d_{\text{Gamma}}^* - \mu_1 = \mu_2 \sigma_1^2 \frac{\mu_1 - \mu_2}{\mu_1 \sigma_2^2 - \mu_2 \sigma_1^2}, \quad d_{\text{Gamma}}^* - \mu_2 = \mu_1 \sigma_2^2 \frac{\mu_1 - \mu_2}{\mu_1 \sigma_2^2 - \mu_2 \sigma_1^2}.$$

These two quantities share the same sign, determined by $\frac{\mu_1 - \mu_2}{\mu_1 \sigma_2^2 - \mu_2 \sigma_1^2}$. Hence,

$$d_{\text{Gamma}}^* \notin (\min\{\mu_1, \mu_2\}, \max\{\mu_1, \mu_2\}).$$

Therefore, the conclusion of Thm. 2 remains unchanged under the Gamma-family model.

C. DARE Instantiation Details

We integrate DARE into two representative offline RL algorithms, Cal-QL (Nakamoto et al., 2023) and IQL (Kostrikov et al., 2022), by applying differentiated updates to the exchanged batches b'_{off} and b'_{on} . For offline-like samples in b'_{off} , we still use the conservative objectives to preserve stability. For online-like samples in b'_{on} , we relax these constraints to facilitate adaptation.

DARE-Cal-QL. We extend Cal-QL with DARE, resulting in the variant DARE-Cal-QL (DARE-C). Cal-QL applies a conservative regularizer $\mathcal{R}$ uniformly to all samples to mitigate value overestimation. In DARE-C, this penalty is applied selectively: $\mathcal{R}$ is retained for offline-like samples in b'_{off} to preserve stability, but removed for online-like samples in b'_{on} to facilitate adaptation. The original objective becomes

$$\begin{aligned} \mathcal{L}_{\text{DARE}}^{\text{Cal-QL}}(Q) = \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \Big[& \mathbb{1}_{\{(s,a,s') \in b'_{off}\}} ((Q(s,a) - \hat{B}^\pi \hat{Q}_{\text{target}}(s,a))^2 + \alpha \mathcal{R}) \\ & + \mathbb{1}_{\{(s,a,s') \in b'_{on}\}} (Q(s,a) - \hat{B}^\pi \hat{Q}_{\text{target}}(s,a))^2 \Big]. \end{aligned} \quad (10)$$

DARE-IQL. When applied to IQL, DARE brings in a variant DARE-IQL (DARE-I). IQL trains the Q-function via value regression and updates the policy through advantage-weighted regression. In DARE-I, these updates are differentiated: for offline-like samples b'_{off} , we retain the original value targets and use the PEX objective for policy learning; for online-like samples b'_{on} , we replace the value targets with TD-based maximum-Q estimates and switch the policy update to an entropy-regularized SAC objective (Haarnoja et al., 2018). The original Q-function and policy updates become:

$$\begin{aligned} \mathcal{L}_{\text{DARE}}^{\text{IQL}}(Q) = \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \Big[& \mathbb{1}_{\{(s,a,s') \in b'_{off}\}} (r(s,a) + \gamma V(s') - Q(s,a))^2 \\ & + \mathbb{1}_{\{(s,a,s') \in b'_{on}\}} (r(s,a) + \gamma \max_{a'} \hat{Q}_{\text{target}}(s',a') - Q(s,a))^2 \Big]. \end{aligned} \quad (11)$$

For policy optimization, the overall policy objective is given by

$$\begin{aligned} \mathcal{L}_{\text{DARE}}^{\text{IQL}}(\pi) = \mathbb{E}_{(s,a) \sim \mathbf{D}, \tilde{a} \sim \tilde{\pi}(s)} \Big[& \mathbb{1}_{\{(s,a) \in b'_{off}\}} w(s,\tilde{a}) \log \pi(\tilde{a} | s) \Big] \\ & + \mathbb{E}_{s \sim \mathbf{D}, \tilde{a} \sim \pi(\cdot | s)} \Big[\mathbb{1}_{\{(s,\tilde{a}) \in b'_{on}\}} (\alpha \log \pi(\tilde{a} | s) - Q(s,\tilde{a})) \Big], \end{aligned} \quad (12)$$

where $\tilde{\pi}$ is the composite policy defined in PEX with $w(s,\tilde{a}) = \exp(\beta(Q(s,\tilde{a}) - V(s)))$, and entropy coefficient α is set to 0.2 for locomotion tasks and 0.02 for AntMaze tasks.

Energy-Guided Diffusion Model as Behavior Policy. Energy-guided diffusion model has been widely deployed to improve data quality or sample efficiency (Lu et al., 2023b; Liu et al., 2024; Zhang et al., 2024b; Xu et al., 2025b). Mathematically, we take the pretrained diffusion behavior model μ_0 as the base distribution and use the Q-function as the energy term, which defines an induced policy π_0 . We have the marginal distributions μ_t and π_t of the noise-perturbed action a_t satisfying:

$$\pi_t(\mathbf{a}_t | \mathbf{s}) \propto \mu_t(\mathbf{a}_t | \mathbf{s}) e^{\mathcal{E}_t(\mathbf{s}, \mathbf{a}_t)}. \quad (13)$$

$\mathcal{E}_t(s, a_t)$ is an intermediate energy function determined by the learned action evaluation model $Q(s, a_0)$. Specifically, $\mathcal{E}_t(s, a_t) = \log \mathbb{E}_{\mu_t(a_0|a_t,s)} [e^{\beta Q(s,a_0)}]$, and $\mathcal{E}_0(s, a_0) = \beta Q(s, a_0)$. To allow action sampling from π_0 , the score function can be expressed as:

$$\nabla_{a_t} \log \pi_t(a_t | s) = \underbrace{\nabla_{a_t} \log \mu_t(a_t | s)}_{\approx -\epsilon_\theta(a_t | s, t) / \sigma_t} + \underbrace{\nabla_{a_t} \mathcal{E}_t(s, a_t)}_{\approx f_\phi(s, a_t, t)}, \quad (14)$$

where $\epsilon_\theta(a_t | s, t) / \sigma_t$ denotes the state-conditional diffusion model to model the behavior policy μ_0 . The energy guidance model f_ϕ employs a contrastive learning objective based on self-normalized energy labels that defined as:

$$\min_{\phi} \mathbb{E}_{p(t)} \mathbb{E}_{q_0(x_0^{(1:K)})} \mathbb{E}_{p(\epsilon^{(1:K)})} \left[- \sum_{i=1}^K \frac{e^{-\beta \mathcal{E}(x_0^{(i)})}}{\sum_{j=1}^K e^{-\beta \mathcal{E}(x_0^{(j)})}} \log \frac{e^{-f_\phi(x_t^{(i)}, t)}}{\sum_{j=1}^K e^{-f_\phi(x_t^{(j)}, t)}} \right]. \quad (15)$$

Here, each perturbed sample is generated by the forward diffusion process: $x_t^{(i)} = \alpha_t x_0^{(i)} + \sigma_t \epsilon^{(i)}$, with constants α_t, σ_t defined by the diffusion schedule.

D. Experimental Details

In our experiments, we evaluate DARE on two standard D4RL benchmarks (Fu et al., 2020): MuJoCo Locomotion and AntMaze Navigation. For instantiation, we extend Cal-QL (Nakamoto et al., 2023) to DARE-C and IQL (Kostrikov et al., 2022) to DARE-I. In the CQL-based group, we compare against CQL (Kumar et al., 2020), Cal-QL, and EDIS-C (Liu et al., 2024), a Cal-QL variant. In the IQL-based group, we include IQL, PEX (Zhang et al., 2023), and EDIS-I, the IQL counterpart of EDIS-C. All models are first trained offline for 1M steps and then fine-tuned online for 0.2M steps. The results are averaged over the last four evaluations and five random seeds. For a fair comparison, **all methods are initialized from the same offline-trained models, using Cal-QL models for EDIS-C and DARE-C and IQL models for PEX, EDIS-I, and DARE-I.**

D.1. Hyperparameters for DARE

DARE introduces only one additional hyperparameter: the update frequency for the class-conditional mean μ_c and variance σ_c of the deviation statistic $d(s, a)$. We update (μ_c, σ_c) every 1000 training steps by sampling 10,000 samples from the offline dataset and 10,000 from the online replay buffer, computing $d(s, a)$ via the reference behavior model, and estimating the statistics separately for each source.

D.2. Hyperparameters for CQL and Cal-QL

We implement the CQL and Cal-QL based on <https://github.com/tinkoff-ai/CORL>, and primarily follow the authors’ recommended hyperparameters (Tarasov et al., 2023). Since CQL-based algorithms are known to be sensitive to hyperparameter choices, we provide the exact settings in our experiments to facilitate the reproducibility. Please refer to Tab. 5 for the details about the hyperparameters in our CQL-based implementation.

Table 5. Hyperparameters in CQL-based implementation.

Hyperparameter	Mojoco locomotion	AntMaze navigation
General Settings		
Replay buffer size	2,000,000	2,000,000
Batch size	256	256
Discount factor γ	0.99	0.99
Reward scale / bias	1.0 / 0.0	10.0 / -5.0
Normalize states	True	False
Normalize reward	False	True
Orthogonal initialization	True	True
Is sparse reward	False	True
CQL Hyperparameters		
Policy learning rate	1×10^{-4}	1×10^{-4}
Critic learning rate	3×10^{-4}	3×10^{-4}
Soft target update rate τ	0.005	0.005
Target update period	1	1
Automatic entropy tuning	True	True
Backup entropy	False	False
CQL regularization (α , offline / online)	10.0 / 10.0	5.0 / 5.0
CQL Lagrange	False	True
CQL temperature	1.0	1.0
Target action gap	-1.0	0.8
Max target backup	False	True
Clip diff range	$[-200, \infty)$	$[-200, \infty)$
Importance sampling	True	True
Network Architecture		
Q-network hidden layers	3	5
Hidden dimension (actor / critic)	256 / 256	256 / 256

D.3. Hyperparameters for IQL and PEX

We implement IQL and PEX based on <https://github.com/Haichao-Zhang/PEX>, the hyperparameters of which is illustrated in Tab. 6.

Table 6. Hyperparameters for the IQL-based experiments.

Hyperparameter	Value
Discount factor γ	0.99
Hidden dimension	256
Number of hidden layers	2
Batch size	256
Learning rate	3×10^{-4}
Target update rate	0.005
Expectile parameter τ	0.9, AntMaze / 0.7, Locomotion
Inverse temperature β	10.0, AntMaze / 3.0, Locomotion

D.4. Hyperparameters for EDIS

The implementation of EDIS are referred to <https://github.com/liuxhym/EDIS>. For EDIS-C, we use its official implementation of Cal-QL. For EDIS-I, we modify the classes of Q-function, value function, and policy function to match those in IQL and PEX, so that the same offline models can be loaded for initialization. The hyperparameters used in the EDIS module remain unchanged and, for convenience, are detailed in Tab. 7.

Table 7. Hyperparameters in EDIS.

Hyperparameter	Value
Network Type (Denoising)	Residual MLP
Denoising Network Depth	6 layers
Denoising Steps	128 steps
Denoising Network Learning Rate	3×10^{-4}
Denoising Network Hidden Dimension	1024 units
Denoising Network Batch Size	256 samples
Denoising Network Activation Function	ReLU
Denoising Network Optimizer	Adam
Learning Rate Schedule (Denoising Network)	Cosine Annealing
Training Epochs (Denoising Network)	50,000 epochs
Training Interval Environment Step (Denoising Network)	Every 10,000 steps
Energy Network Hidden Dimension	256 units
Negative Samples (Energy Network Training)	10
Energy Network Learning Rate	1×10^{-3}
Energy Network Activation Function	ReLU
Energy Network Optimizer	Adam

The guidance scale s follows the empirical settings from prior work (Lu et al., 2023b), as summarized in Tab. 8.

Table 8. Guidance scale s across different environments.

Dataset	Guidance Scale s
walker2d-medium-expert-v2	5.0
halfcheetah-medium-expert-v2	3.0
hopper-medium-expert-v2	2.0
walker2d-medium-replay-v2	5.0
halfcheetah-medium-replay-v2	8.0
hopper-medium-replay-v2	3.0
walker2d-medium-v2	10.0
halfcheetah-medium-v2	10.0
hopper-medium-v2	8.0
antmaze-umaze-v2	3.0
antmaze-medium-play-v2	4.0
antmaze-large-play-v2	2.0
antmaze-umaze-diverse-v2	1.0
antmaze-medium-diverse-v2	3.0
antmaze-large-diverse-v2	3.0

D.5. Offline Model Performance

Before fine-tuning, we present the performance of the offline-trained models on both Locomotion and AntMaze benchmarks. These results serve as the initialization for all compared methods, ensuring that performance differences in the online phase come solely from the fine-tuning strategy rather than the quality of the offline implementation codebases. Tab. 9 summarizes the scores of CQL, Cal-QL and IQL across different datasets.

Table 9. Offline training performance before online fine-tuning.

Dataset	CQL	Cal-QL	IQL
HC-ME	90.4 $\pm$ 3.5	82.9 $\pm$ 4.1	90.2 $\pm$ 1.4
H-ME	108.1 $\pm$ 2.4	109.0 $\pm$ 2.1	32.8 $\pm$ 32.6
W2D-ME	109.7 $\pm$ 0.2	108.6 $\pm$ 0.8	108.3 $\pm$ 2.0
HC-MR	45.5 $\pm$ 0.2	46.5 $\pm$ 0.2	43.9 $\pm$ 0.4
H-MR	90.2 $\pm$ 10.5	64.8 $\pm$ 27.1	84.2 $\pm$ 13.6
W2D-MR	78.6 $\pm$ 4.5	85.0 $\pm$ 2.9	70.8 $\pm$ 4.1
HC-M	47.1 $\pm$ 0.3	48.3 $\pm$ 0.3	48.1 $\pm$ 0.1
H-M	63.9 $\pm$ 1.5	61.9 $\pm$ 2.0	56.2 $\pm$ 4.4
W2D-M	81.9 $\pm$ 1.2	83.7 $\pm$ 1.2	74.0 $\pm$ 3.1
total (L)	715.4	690.7	608.5
AM-LD	30.6 $\pm$ 5.6	34.8 $\pm$ 13.4	35.2 $\pm$ 10.0
AM-LP	32.7 $\pm$ 4.1	42.4 $\pm$ 9.7	36.6 $\pm$ 3.2
AM-MD	57.6 $\pm$ 3.5	66.2 $\pm$ 6.1	70.4 $\pm$ 5.2
AM-MP	64.8 $\pm$ 6.1	68.4 $\pm$ 3.8	74.6 $\pm$ 4.7
AM-UD	28.2 $\pm$ 23.9	46.0 $\pm$ 20.1	60.2 $\pm$ 6.4
AM-U	87.6 $\pm$ 4.6	76.6 $\pm$ 2.7	91.4 $\pm$ 0.8
total (AM)	301.5	334.4	368.4
total	1016.9	1025.1	976.9

E. Comparisons of Online Fine-tuning Processes

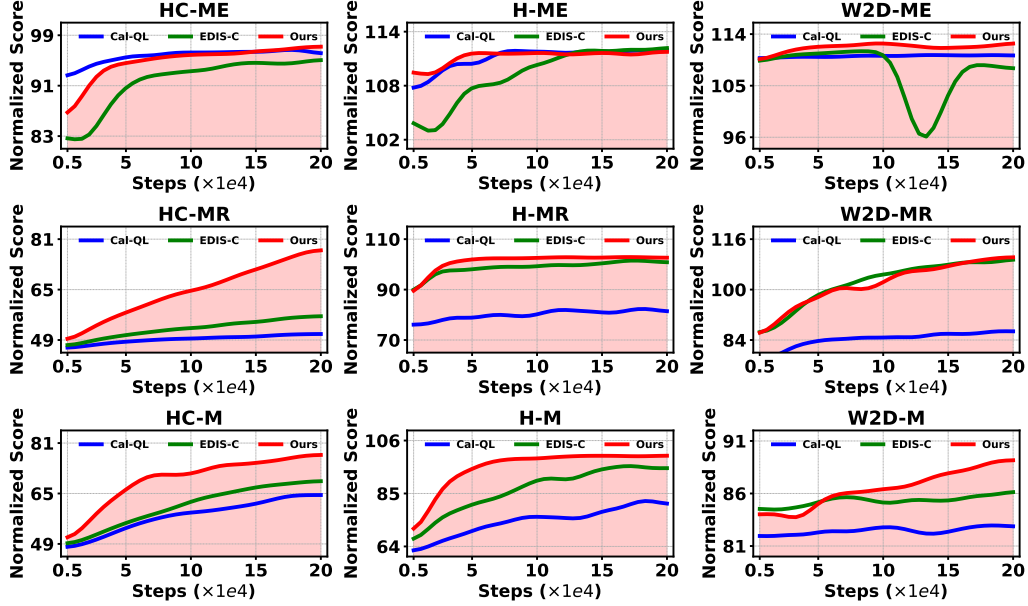


Figure 7. Online training curves on Locomotion benchmarks comparing Cal-QL, EDIS-C, and Ours.

Complementary to the main results in the paper, we depict the online training dynamics across algorithms. As shown in Figs. 7 our method consistently outperforms both Cal-QL and EDIS-C, while exhibiting smoother training trajectories.

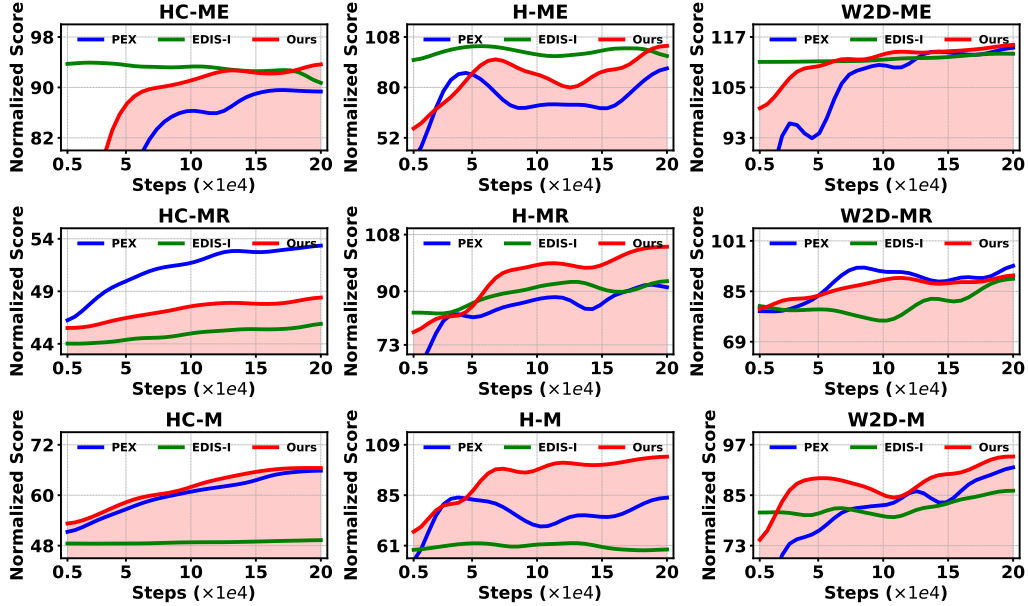


Figure 8. Online training curves on Locomotion benchmarks comparing PEX, EDIS-I, and Ours.

As shown in Figs. 8, our method consistently exhibits smoother optimization dynamics and higher asymptotic returns than both PEX and EDIS-I across most benchmarks, with the exceptions of *halfcheetah-medium-replay* and *walker2d-medium-replay*. The curves demonstrate that EDIS-I attains strong initial performance on expert datasets and effectively exploits high-quality offline data at early stages. However, its performance declines as the optimization process continues, which suggests limited robustness during the online adaptation phase.

F. Additional Results for Main Table

This section provides additional experimental results to complement the main results reported in Tab. 1.

Results on Kitchen tasks. We evaluate DARE on the Kitchen tasks (Gupta et al., 2020) under the IQL-based setting. As shown in Tab. 10, DARE-I consistently outperforms both IQL and PEX, which supports the effectiveness of DARE.

Table 10. Results on Kitchen tasks under the IQL-based setting with 0.5M offline training steps and 1M online fine-tuning steps.

Task	IQL	PEX	DARE-I
kitchen-partial-v0	42.29	44.17	50.21
kitchen-mixed-v0	53.12	53.75	56.25

Results on different instantiations. We examine the generality of the exchange mechanism on the Locomotion tasks under different online-objective instantiations. In these experiments, the exchange mechanism is kept fixed, while the online objective applied to online-like samples is replaced with different alternatives. As shown in Tab. 11, enabling exchange consistently improves performance over the corresponding no-exchange variants across all instantiations, demonstrating that the proposed mechanism is not tied to a specific online objective.

Table 11. Locomotion results for different instantiations of DARE-I. w/o S (no SAC), w/o P (no PEX), w/o S&P (TD-only), with w/o E. and w/ E. indicating without and with exchange.

M: medium	DARE-I.w/o S		DARE-I.w/o P		DARE-I.w/o S&P	
MR: medium-replay	w/o E.	w/ E.	w/o E.	w/ E.	w/o E.	w/ E.
M(total)	231.1	238.4	207.2	230.8	232.4	247.9
MR(total)	259.2	261.4	215.4	227.0	223.3	229.5

Results on the TD3+BC backbone. The main results in Tab. 1 are primarily based on CQL and IQL backbones. To further examine the generality of DARE, we extend the proposed exchange mechanism to TD3+BC (Fujimoto & Gu, 2021), another representative offline RL algorithm that regularizes the TD3 actor update with a behavior cloning (BC) term

$$\pi = \arg \max_{\pi} \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[\lambda Q(s, \pi(s)) - \|\pi(s) - a\|_2^2 \right],$$

where λ is a weighting coefficient that balances Q-value maximization and behavior cloning regularization.

For evaluation, we apply DARE to two TD3+BC-based O2O methods, BOORL (Hu et al., 2024) and LAROO (Guo et al., 2026). During online fine-tuning, the offline objective active by applying it once every two gradient updates, so that the actor remains regularized by identified offline data via DARE. In addition, although the official implementations of BOORL and LAROO use a UTD ratio of 5, we set the UTD ratio to 1 in our experiments to match the training setting used throughout this paper and ensure a fair comparison. As shown in Tab. 12, DARE improves the corresponding baselines, further indicating that the proposed mechanism can generalize beyond CQL- and IQL-based settings.

Table 12. Locomotion results for.

	BOORL		LAROO	
	base	+DARE	base	+DARE
HC-M	91.9	93.9	92.5	94.1
HC-MR	82.6	84.1	80.2	78.5
H-M	108.1	108.2	109.0	110.0
H-MR	104.2	109.8	107.9	108.6
W2D-M	97.6	100.9	106.9	111.9
W2D-MR	103.1	108.1	105.6	109.1

G. Visualization of Exchange Behavior

We present the exchange behavior of DARE-C and DARE-I during online training across all tested environments. We depict the number of exchanges over training iterations to characterize how often the proposed exchange mechanism triggers sample reassignment in practice, as shown in Fig. 9.

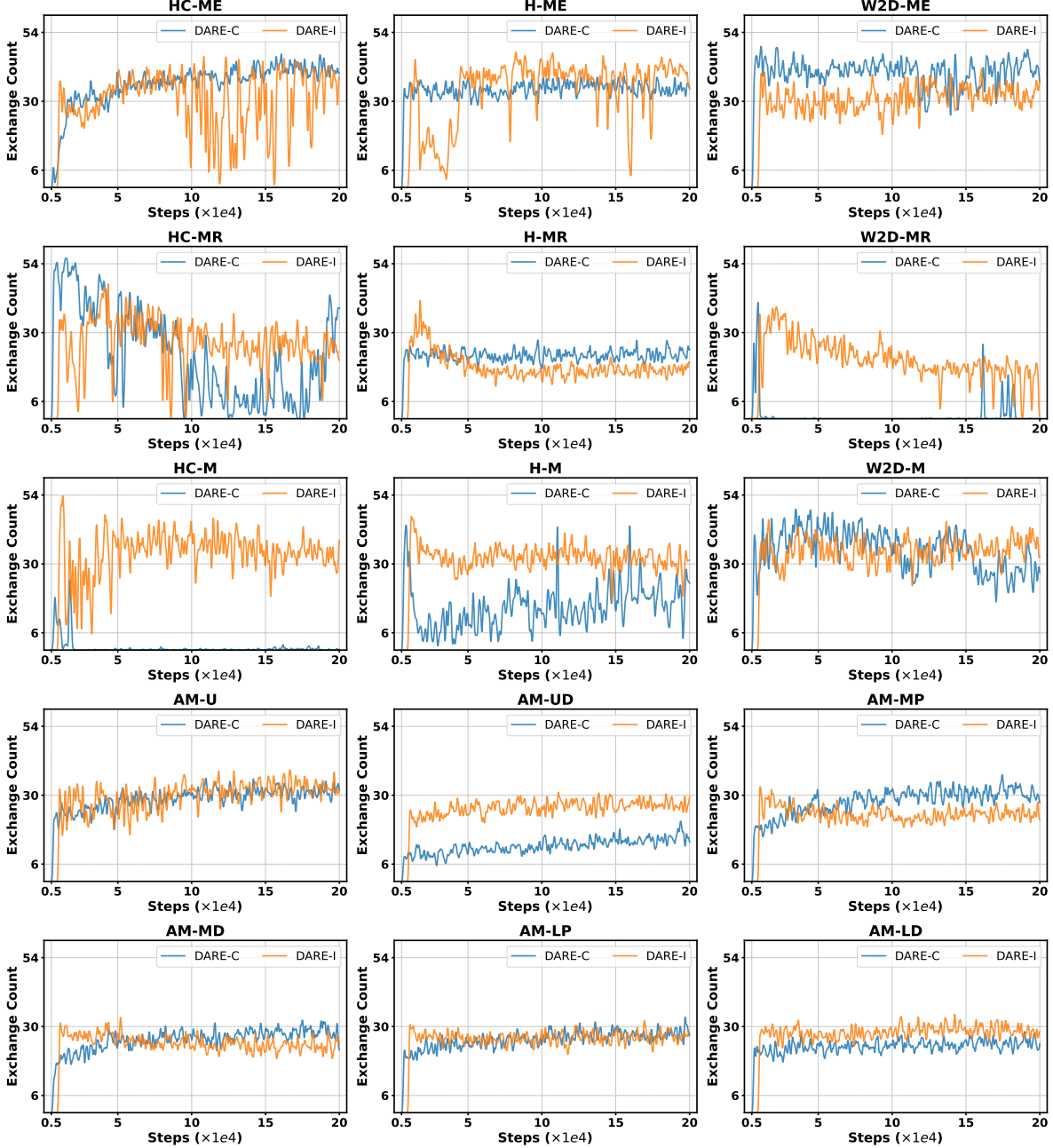


Figure 9. Adaptive exchange behavior of DARE-C and DARE-I during online training across different tasks.

H. Further Results and Analysis on Adroit

We study how different online update schemes interact with Adroit manipulation tasks under a fixed exchange mechanism. PEX constrains online interaction at the behavior level rather than through an explicit objective. Following Zhang et al. (2023), the behavior policy is constructed by composing the frozen offline policy π_β and the online policy π_θ , with actions selected according to a critic-induced distribution $P(i | s) \propto \exp(Q(s, a_i)/\alpha)$, $a_i \sim \{\pi_\beta(s), \pi_\theta(s)\}$. As a result, executed actions remain restricted to proposals from π_β and π_θ . When the offline initialization is weak, this restriction limits deviation from suboptimal contact behaviors in Adroit tasks.

Entropy-regularized SAC objectives exhibit different characteristics. Adroit tasks require fine-grained, contact-rich control, and high action entropy does not consistently align with these requirements (Fu et al., 2020). At the same time, SAC promotes deviation from offline actions and introduces exploratory behavior that is absent from PEX-based updates, which can be beneficial when offline coverage is poor.

Taken together, these results indicate that the effectiveness of online objective instantiations depends on task characteristics and the quality of offline initialization. DARE therefore adopts a objective-agnostic exchange mechanism, which allows different online objectives to be instantiated without modifying the underlying sample reassignment rule. In Fig. 10, the exchange-based method exhibits stronger overall performance.

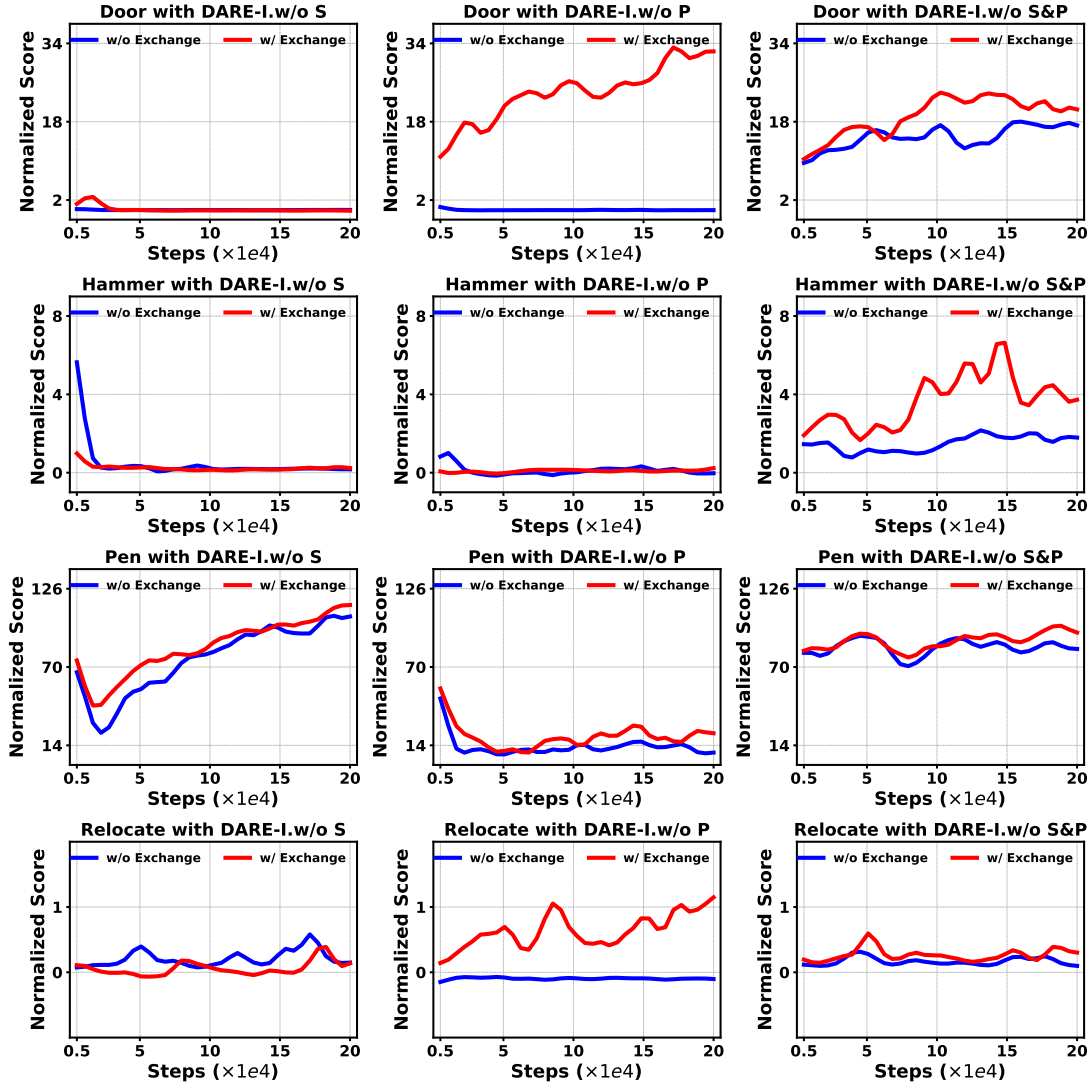


Figure 10. Training curves on Adroit manipulation tasks.

I. Alternative Behavior Model Instantiations

We compare two behavior model instantiations in DARE. We consider a diffusion-based behavior model trained on the offline dataset and an alternative instantiation that uses the online policy actor as a consistency model for online behaviors. Fig. 11 and Fig. 12 present the fine-tuning curves for DARE-I and DARE-C under these two behavior model choices. In most cases, the diffusion-based behavior model achieves slightly better performance. An exception arises in the AntMaze environments under DARE-C, where the policy-based behavior model leads to improved policy performance compared to the diffusion-based instantiation. Tab. 13 further summarizes the overall performance of the two behavior model instantiations across task categories.

Table 13. Comparison of policy-based and diffusion-based behavior model instantiations

	DARE-C		DARE-I	
	Policy	Diffusion	Policy	Diffusion
Locomotion (total)	865.0	878.1	812.1	819.8
AntMaze (total)	539.8	537.3	417.1	433.2
Total	1404.8	1415.4	1229.2	1253.0

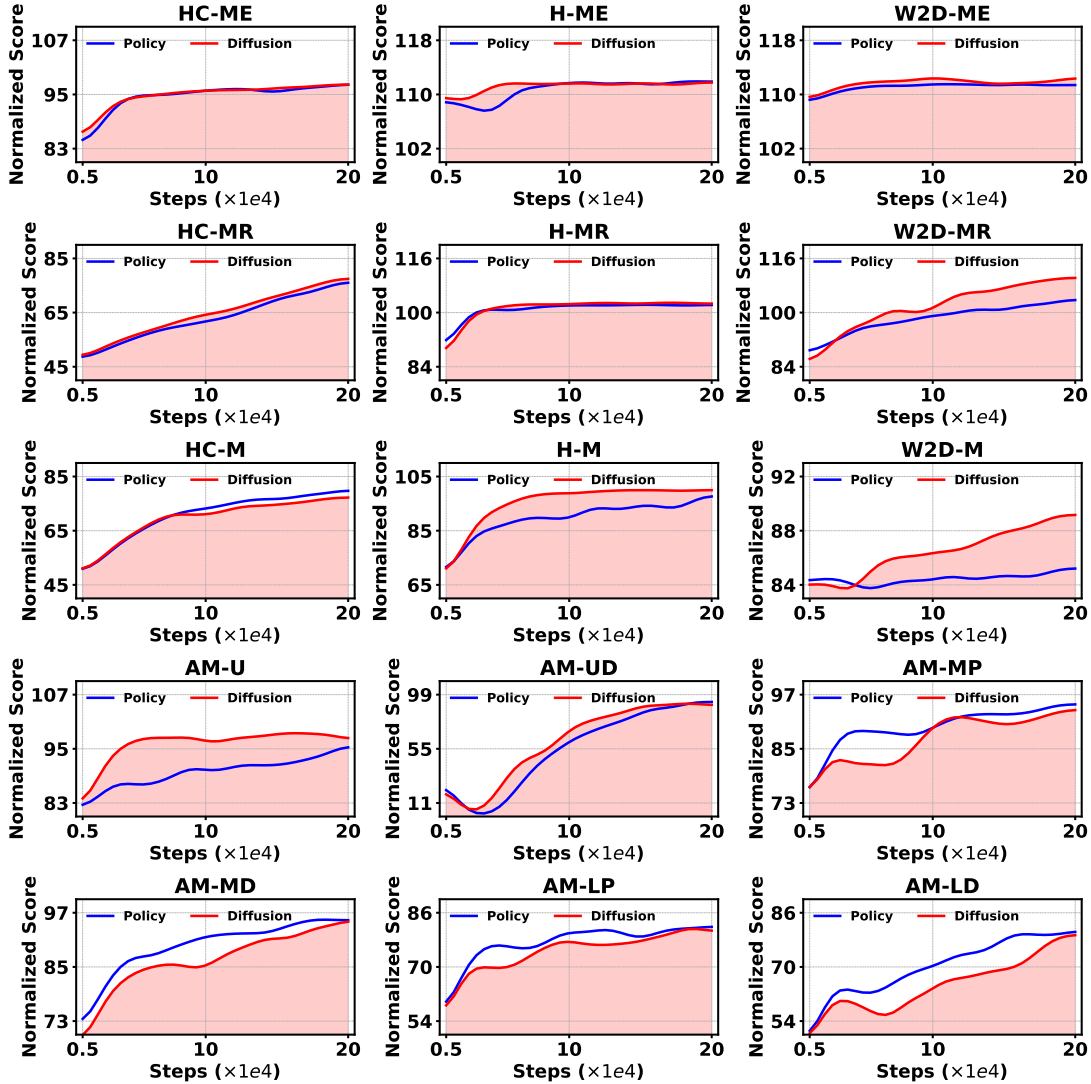


Figure 11. Fine-tuning curves with different behavior model instantiations for DARE-C.

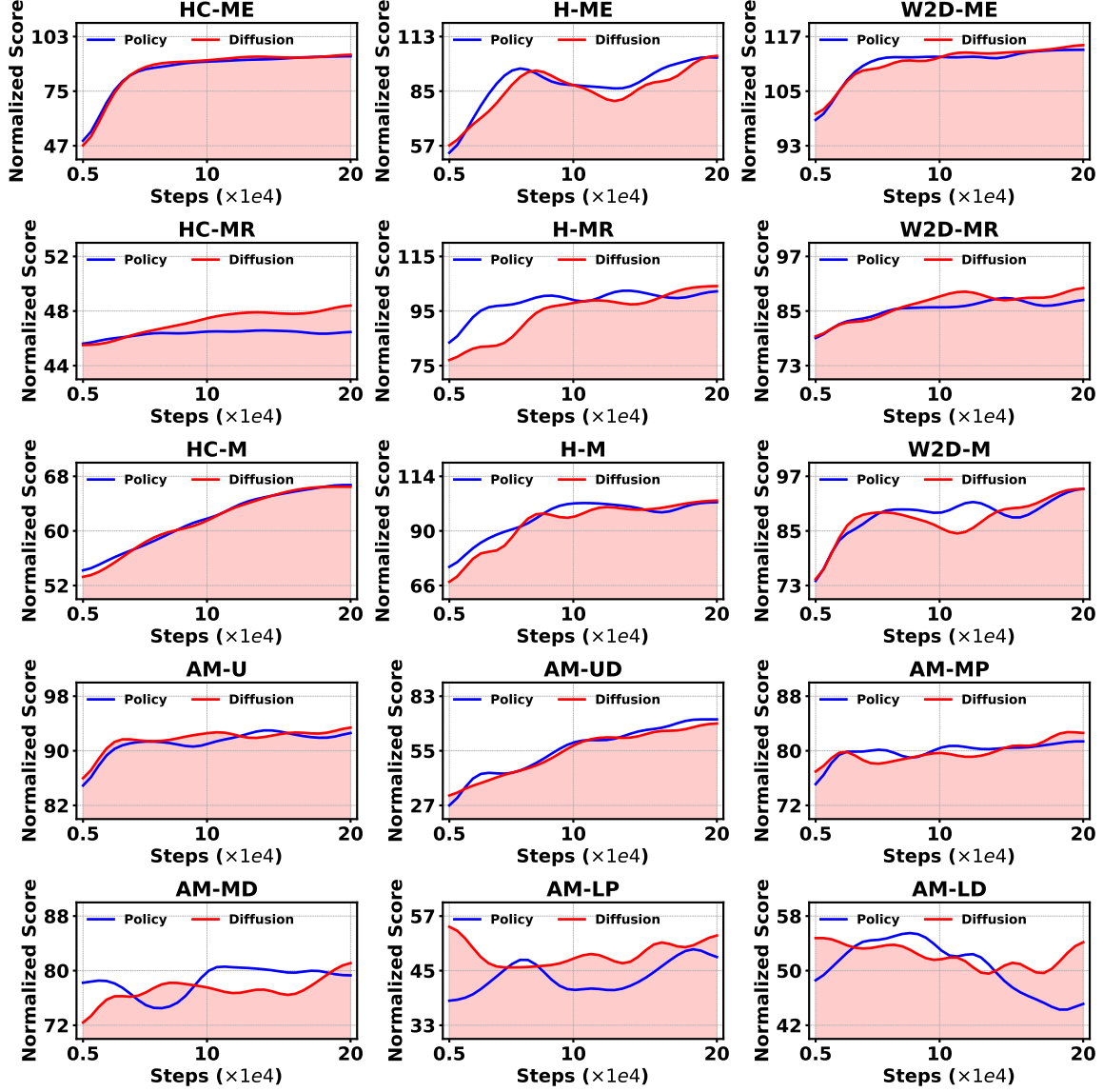
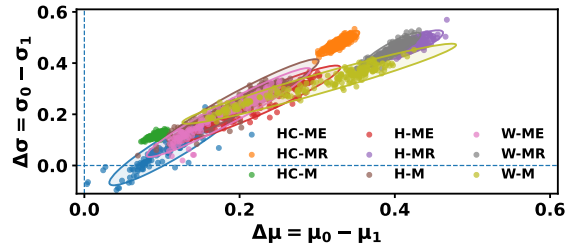


Figure 12. Fine-tuning curves with different behavior model instantiations for DARE-I.

We emphasize that DARE does not restrict the behavior model to a specific instantiation and allows alternative choices based on practical considerations. We also show the evolution of the estimated Gaussian statistics in the policy-based case, as shown in Fig. 13.


 Figure 13. Estimated μ and σ gaps during $0.2M$ training steps, both computed using the ongoing policy that represents for online side.

J. Additional Results for Reversed Behavior Assignment

We illustrate the fine-tuning performance curves for DARE-I.C (curiosity-style) and the standard DARE-I under the diffusion-based behavior model in Fig. 14 and the policy-based behavior model in Fig. 15. Across all MuJoCo locomotion tasks, the reversed assignment (DARE-I.C) is consistently worse than the standard DARE-I, indicating that the inverse exchange operation degrades fine-tuning performance.

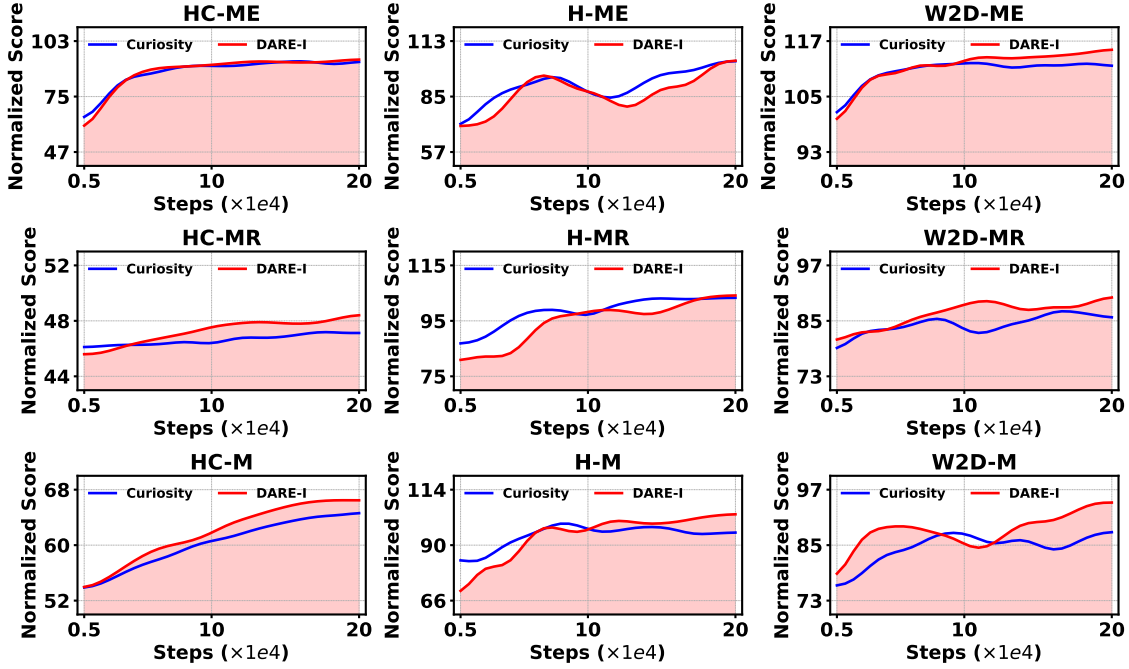


Figure 14. Comparison of fine-tuning curves between DARE-I.C (curiosity-style) and DARE-I using a diffusion-based behavior model.

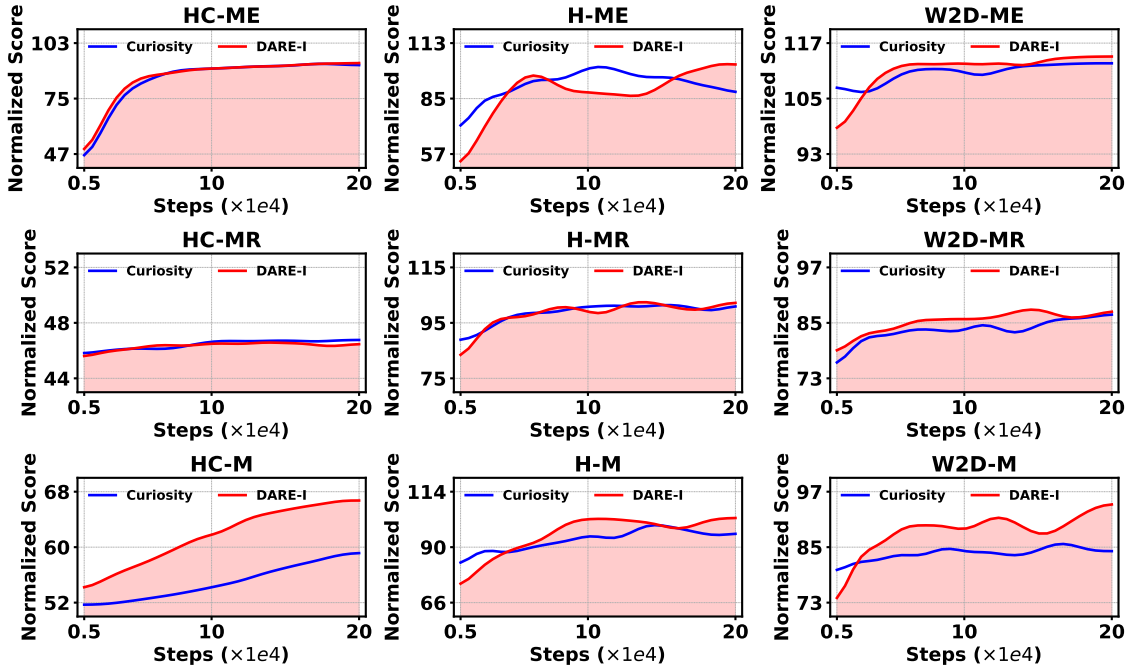


Figure 15. Comparison of fine-tuning curves between DARE-I.C (curiosity-style) and DARE-I using a policy-based behavior model.

K. Additional Ablation Study

Ablation learning curves. In Fig. 16, we visualize the training trajectories of DARE-I and its w/o Exchange ablation to illustrate the effect of removing the exchange mechanism during training.

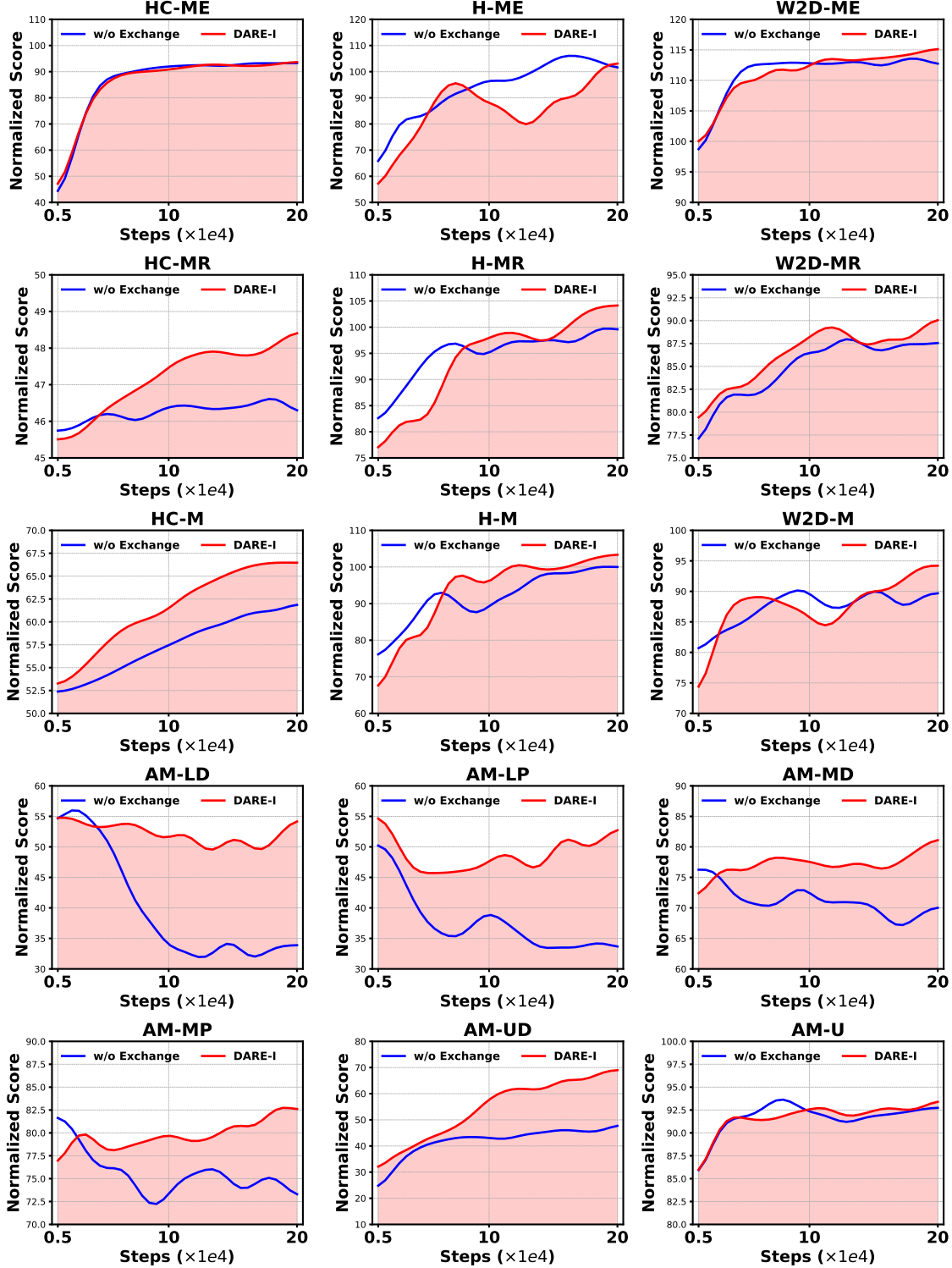


Figure 16. Performance curves comparing DARE-I with and without the exchange mechanism.

Maximum number of exchanges. As described in Eq. 8, DARE swaps the same number of samples in the two directions to keep the exchange operation balanced

$$K = \min(|\mathcal{P}_{\text{off} \rightarrow \text{on}}|, |\mathcal{P}_{\text{on} \rightarrow \text{off}}|).$$

In implementation, we optionally impose an upper bound on the number of exchanged samples

$$K = \min(|\mathcal{P}_{\text{off} \rightarrow \text{on}}|, |\mathcal{P}_{\text{on} \rightarrow \text{off}}|, K_{\max}).$$

This option controls the maximum number of exchanged samples in each mini-batch and prevents unusually large exchanges from dominating a single update. We further examine the effect of this cap on Locomotion tasks. Setting $K_{\max} = 32$ achieves a total score of 877.1, while setting $K_{\max} = 128$, which is effectively uncapped under our batch setting, achieves 878.2. This observation is also consistent with Fig. 9, where the number of exchanged samples typically stays around 30.

Behavioral distance choices. In DARE, the L_2 distance is used as a scalar behavioral deviation statistic between the action a and the behavior reference $a_b(s)$, rather than as a full probabilistic discrepancy measure. For continuous-control O2O RL, L_2 provides a simple and parameter-free measure of action deviation, which is sufficient for sample ordering and posterior-based assignment. This choice also keeps the evaluation focused on the contribution of DARE itself, rather than introducing additional complexity through metric engineering. Nevertheless, we further compare DARE-I with alternative distance measures, including L_1 distance and Mahalanobis distance. As shown in Tab. 14, the results suggest that DARE is not overly sensitive to the specific behavioral distance measure used in DARE.

Table 14. Results of DARE-I with different behavioral distance measures.

Locomotion	L_1	Mahalanobis	L_2 (Ours)
M (total)	267.3	261.9	264.5
MR (total)	238.5	237.1	242.3

Runtime comparison. We further evaluate the runtime on `hopper-medium-replay-v2`, with all experiments conducted on the same NVIDIA GeForce RTX 5090 GPU. As shown in Tab. 15, DARE-C (Policy) incurs only a small overhead compared with Cal-QL, while being substantially faster than EDIS-C. DARE-C (Diffusion) requires additional computation due to its diffusion-based behavior modeling component. Since DARE does not require the behavior reference to be parameterized by a diffusion model, the policy-based variant provides a more computationally efficient instantiation when runtime overhead is a concern.

Table 15. Runtime comparison on `hopper-medium-replay-v2`.

	Cal-QL	EDIS-C	DARE-C (Diffusion)	DARE-C (Policy)
Runtime	0.5h	1.6h	2.1h	0.6h